

BIOLOGY GRADE 10: CHEMICALS OF LIFE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.2 (6 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Biology
Strand	Strand 1.0: Cell Biology and Biodiversity
Sub-Strand	Sub-Strand 1.2: Chemicals of Life
Total Duration	6 lessons × 40 minutes = 240 minutes total
Sub-Strand Content	– Elements and compounds found in living organisms – Organic and inorganic compounds in cells – Carbohydrates: structure, types (monosaccharides, disaccharides, polysaccharides) and functions – Lipids: structure, types and functions – Proteins: structure, amino acids and functions – Water and mineral salts: properties and roles in living organisms – Food tests: starch, reducing and non-reducing sugars, proteins, lipids – Enzymes as biological catalysts in the breakdown of food molecules – Release of energy from food molecules in cells and regulation of blood glucose
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) identify the elements and compounds that make up living organisms, b) describe the structure and functions of carbohydrates, lipids and proteins, c) carry out food tests to identify nutrients in common Kenyan foods, d) explain the roles of water, mineral salts and enzymes in living organisms, e) appreciate the importance of chemicals of life in nutrition and health.
Core Competencies	– Critical Thinking and Problem Solving: Learners analyse the blood glucose graph and food test results to explain how food chemicals are processed in the body. – Communication and Collaboration: Learners work in groups to carry out food tests, share findings and revise a class model. – Learning to Learn: Learners track their growing understanding using the Driving Question Board and revised models. – Digital Literacy: Learners interpret graphs and use simulations or videos on digestion and glucose regulation. – Self-efficacy: Learners build confidence making and defending evidence-based predictions about their own diets.
Core Values	– Responsibility: Learners handle chemicals and food-test apparatus safely and make responsible food choices. – Respect: Learners value different ideas and dietary traditions during discussions. – Unity: Learners cooperate in groups to collect and share evidence. – Integrity: Learners report experimental results honestly even when they differ from predictions.

Science & Engineering Practices	– Asking Questions: Learners generate questions for the DQB from the glucose graph. – Developing and Using Models: Learners build and revise a model of how food chemicals are digested, absorbed and used by cells. – Planning and Carrying Out Investigations: Learners test foods such as ugali, githeri, mandazi, milk and groundnuts for nutrients. – Analysing and Interpreting Data: Learners interpret the blood glucose-time graph and food test results. – Constructing Explanations: Learners explain why blood glucose rises then falls after a meal.
Pertinent & Contemporary Issues (PCIs)	– Health Education: Learners link nutrient composition of foods to balanced diets and the prevention of diabetes and malnutrition. – Safety and Security: Learners use chemicals such as iodine, Benedict's and biuret solutions safely. – Environmental Conservation: Learners use locally available food materials and dispose of waste responsibly.
Career Connections	– Nutritionist/Dietitian: applies knowledge of food chemicals to plan balanced meals. – Medical Laboratory Technologist: tests blood glucose and analyses samples. – Doctor/Clinical Officer: diagnoses and manages diabetes and nutritional disorders. – Food Technologist: tests and processes food at companies in Nairobi and Kisumu. – Agricultural Extension Officer: advises farmers in the Rift Valley on nutritious crops such as maize and beans.
Focus for Lessons	This unit uses the puzzling rise and fall of blood glucose after a meal to drive investigation into the chemicals in food (carbohydrates, proteins, lipids, water and minerals) and how enzymes and cells process them. Learners predict, test common Kenyan foods, make sense of evidence, update the Driving Question Board and progressively revise a model of what happens to food chemicals in the body.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: What chemicals are in the food we eat, and what happens to them in our bodies? KICD KEY INQUIRY QUESTIONS: 1. What are the chemicals of life? 2. How do the chemicals of life contribute to the functioning of living organisms?
Anchoring Phenomenon	After eating a meal, for example a plate of ugali, sukuma wiki and beans in Nairobi, the glucose concentration in a person's blood rises sharply and then gradually falls back to normal. Learners are shown a graph of blood glucose concentration over time after eating: a spike soon after the meal followed by a gradual decrease to the starting level. Why does the concentration of glucose in your blood rise and then fall after eating a meal? What chemicals in the food you ate caused this change, and how do your cells process these chemicals? This is puzzling because students eat every day but may not know what happens chemically. The graph shows a pattern, so something is controlling glucose levels. What chemicals are in food? How do cells break them down? Why doesn't glucose stay high forever? What is happening inside cells to process these chemicals?
Storyline Thread	Lesson 1: Students view the blood glucose graph after a meal, predict what causes the rise and fall, and create the first Driving Question Board questions and an initial model. Lesson 2: Students identify the elements and compounds in living things and classify nutrients in Kenyan foods (ugali, beans, sukuma wiki, milk, nyama choma) as organic or inorganic. Lesson 3: Students investigate carbohydrates by testing foods for starch and reducing sugars, linking sugars from ugali and maize to the glucose spike, and revise the model. Lesson 4: Students investigate proteins and lipids with biuret and emulsion or grease-spot tests on beans, groundnuts, milk and mandazi, and compare their roles and structures with carbohydrates.

	Lesson 5: Students explore the roles of water, mineral salts and enzymes in breaking down starch to glucose, using the saliva and starch investigation to explain how food chemicals are processed. Lesson 6: Students explain how glucose is absorbed, used in cell respiration for energy and regulated back to normal levels, update the DQB, and finalise and present their model answering the driving question.
Supporting Phenomena	– A long-distance runner from Iten eats githeri and ugali the night before a race and has sustained energy the next day. – A pot of boiled maize flour porridge (uji) turns thinner when saliva is added, because enzymes break down starch. – Groundnuts or mandazi leave a grease spot on paper, showing that lipids are present in foods. – A person with diabetes in Kisumu must monitor blood glucose after meals because the glucose level does not return to normal as expected.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.

LESSON 1 (80 minutes (double lesson: 2 x 40 minutes)): The Blood Glucose Puzzle: Why Does Glucose Rise and Fall After Eating Ugali?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners launch the anchoring phenomenon by reading the blood glucose graph of Wanjiru, a Grade 10 learner in Nairobi, after a meal of ugali, sukuma wiki and beans. They make predictions, describe the spike-then-fall pattern with numbers, and ask testable questions about which chemicals in food and which processes cause it.
Knowledge	Learners describe the pattern in the data: blood glucose concentration rises from 4.5 mmol/L to a peak of 8.0 mmol/L at 60 minutes (an increase of 3.5 mmol/L, about 78% above the starting level), then falls back to 4.5 mmol/L by 180 minutes (a decrease of 3.5 mmol/L over 120 minutes). They recognise that the rise is faster than the fall, that the meal contains several kinds of food chemicals (about 70 g carbohydrate in ugali, about 20 g protein and 15 g carbohydrate in beans, about 3 g protein plus fibre, vitamins and minerals in sukuma wiki, about 10 g fat from cooking oil), and that the pattern suggests something in the body is controlling glucose. KICD Key Inquiry Questions: What are the chemicals of life? How do the chemicals of life contribute to the functioning of living organisms?
Skills	Analysing and interpreting data (reading a line graph and a table, calculating change and percentage change); asking questions that can be investigated; developing and using an initial model (a flow diagram of food to blood glucose to normal); constructing explanations from evidence; communicating ideas in pairs and as a whole class. Core competencies: Critical thinking and problem solving; Communication and collaboration; Learning to learn.
Attitudes	Curiosity about everyday phenomena such as eating a meal; respect for peers' ideas, including incomplete ones; willingness to say 'I do not know yet' and to revise ideas with evidence; appreciation of the importance of Biology in everyday life, including the food we eat.
Key Inquiry Question	What chemicals are in the food we eat, and what happens to them in our bodies? (Sub-questions: What are the chemicals of life? How do the chemicals of life contribute to the functioning of living organisms?)
Purpose in Storyline	Lesson 1 opens the storyline. Learners meet the puzzle, since they eat ugali, sukuma wiki and beans every day but cannot yet explain the chemistry. They record first ideas, create the first questions on the Driving Question Board (DQB), and draw an initial model. These first questions and the model will be revisited and revised in every later lesson, starting with Lesson 2, where learners find out what elements and compounds make up the meal's chemicals.
Safety Notes	No chemicals or heating in this lesson. Do not ask learners to test their own blood glucose or to fast. If any learner has diabetes or a family member who does, handle the topic sensitively, invite but never require personal sharing, and avoid giving medical advice. Remind learners that Wanjiru's graph is data for one adult and is not a diagnosis. Use clean hands if sharing real food items for a display, and check for food allergies (for example groundnuts) before bringing any food into class.

B. LESSON OVERVIEW

Mr Otieno, a Biology teacher in Nairobi, tells his class that his learner Wanjiru ate lunch of ugali (300 g), sukuma wiki (100 g) and beans (100 g) between 0 and 15 minutes, after fasting, and that her blood glucose was measured for 3 hours. Learners predict the shape of the graph before seeing it, then read the real data (table and graph): 4.5, 5.5, 7.0, 8.0 (peak), 6.5, 5.2, 4.7, 4.5 mmol/L at 0, 15, 30, 60, 90, 120, 150 and 180 min. They calculate the rise (3.5 mmol/L in 60 min, about 78% above the start) and the fall (3.5 mmol/L over 120 min), notice that the rise is faster than the fall, and note that the level is back in the normal fasting range of 4.0 to 5.5 mmol/L by 150 min. Using a Notice-Wonder-Question routine, they generate questions about the chemicals in the meal and what the body does with them. The class sorts questions into categories on the first DQB, and each learner draws an initial flow-diagram model (food chemicals, then glucose in blood, then back to normal) with

question marks where they are unsure. The lesson ends with a Summary Table (We observed, We learned, We can explain) and a reminder that 'You didn't just learn ABOUT science, you DID science!'

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Reading line graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: chemicals life blood glucose	Learners sit in pairs. Mr Otieno says Wanjiru, a Grade 10 learner in Nairobi, ate ugali, sukuma wiki and beans after fasting, and her blood glucose concentration was measured at the start and for 3 hours after. Each learner individually sketches a prediction graph in their notebook: x-axis time (0 to 180 min), y-axis blood glucose concentration (mmol/L), with the starting value marked as 4.5 mmol/L. Under the sketch they write one sentence: 'I think the line will ... because ...'. Pairs compare sketches and agree on one reason for the shape. A few pairs hang their predictions on the wall or board. The class takes a quick hand vote on three shapes: stays flat, goes up and stays up, goes up then comes down.	Hold up a plate or picture of ugali, sukuma wiki and beans and ask: 'Who ate something like this in the last two days? Raise your hand.' Say: 'Today we are going to be scientists solving a puzzle that happens inside you after every meal.' Give instructions clearly: 'Work alone for 3 minutes. Draw what you think happens to blood glucose after Wanjiru's lunch.' Circulate and do not correct sketches; say only 'Tell me more about why you drew it that way.' Ask: 'What do you think glucose is, and where in the meal might it come from?' Then WAIT TIME of 10-15 seconds before taking answers. Cold-call 3 learners (one from each vote group) to explain their reason, using 'Thank you, who has a different idea?' after each. Do not say which shape is correct. Tell learners: 'Keep your sketch. We will compare it with real data in a moment.'	Prediction with reasoning (Predict-Observe-Explain). Writing a prediction and a reason before seeing data makes learners commit to prior ideas, and it exposes everyday ideas, such as 'glucose just goes up and stays up', that real data can then confirm or challenge. This creates the need to know that drives the unit.	Look for: (1) each learner's sketch has labelled axes with units (mmol/L, minutes) and a starting value near 4.5; (2) the written reason refers to food or the body, not just guessing; (3) at least 3 different reasons are voiced during cold-calls. Note on a checklist which learners predict 'up then down' and which predict 'up and stay up' or 'flat'. Note any ideas that glucose comes only from sugary foods or that ugali has no connection to glucose. These will be revisited in later lessons.
Observe Phase  VIDEO: Reading line graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely</i>	Learners receive the data table and graph of Wanjiru's blood glucose. Table (time after meal start in min: glucose in mmol/L): 0: 4.5; 15: 5.5; 30: 7.0; 60: 8.0 (peak); 90: 6.5; 120: 5.2; 150: 4.7; 180: 4.5. The meal is described as ugali 300 g, sukuma wiki 100 g and beans 100 g eaten between 0 and 15 min. In pairs they (a) plot or trace the graph on graph paper if only the table is available, (b) use	Distribute the table and graph. Say: 'This is real-pattern data from one adult in Nairobi who had not eaten before the meal. Look at it silently for 2 minutes. Do not talk yet.' Then: 'Write 3 things you notice and 2 things you wonder.' Circulate and ask: 'Which number tells you the peak? How do you know it is the highest?' and 'How could we work out how much it rose?' Support weaker learners to	Notice-Wonder with data analysis (Analysing and Interpreting Data). Separating what the data show (notice) from what we wonder about (questions) trains learners to ground explanations in evidence. Calculating change and rate (3.5 mmol/L in 60 min versus 120 min) turns a visual pattern into quantitative evidence that something is actively regulating glucose.	Check that learners can: (1) read the peak (8.0 mmol/L at 60 min) from the table or graph; (2) correctly calculate the rise (3.5 mmol/L) and the fall (3.5 mmol/L); (3) state that the rise is faster than the fall using numbers; (4) identify that glucose returned to the fasting range (4.0 to 5.5 mmol/L) by 150 min; (5) write at least 2 'wonder' statements. Common errors to listen for: reading the graph as if

<i>matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: chemicals life blood glucose	a Notice-Wonder routine, with each pair writing 3 things they notice and 2 things they wonder on sticky notes, (c) answer data questions: What was the highest value and when? By how much did glucose rise from 0 to 60 min? (3.5 mmol/L, about 78% above the start) By how much did it fall from 60 to 180 min? (3.5 mmol/L) How long did the rise take and how long the fall? (60 min and 120 min). Which is faster, rise or fall? (Rise: 3.5 mmol/L in 60 min compared with 3.5 mmol/L over 120 min). When did glucose return to the normal fasting range of 4.0 to 5.5 mmol/L? (by 150 min at 4.7). Learners then look at the meal composition card: ugali about 70 g carbohydrate (mostly starch from maize); beans about 20 g protein and 15 g carbohydrate; sukuma wiki about 3 g protein plus fibre, vitamins A and C and mineral salts (iron, calcium); about 10 g fat from cooking oil. They compare their predictions with the real graph and mark where they were right and where they were surprised.	subtract 8.0 minus 4.5 and to calculate 3.5 divided by 4.5 to get about 78%. Ask: 'Is the rise or the fall quicker? Prove it with numbers.' WAIT TIME of 10-15 seconds before taking answers. Cold-call 4 pairs to share one noticing each, recording it on the board under 'We notice'. Then show the meal card and ask: 'Which foods in the meal do you think could affect glucose most? Why?' Cold-call 3 learners. Do not confirm yet. Say: 'Hold that thought, as that is a question we will investigate.' Ask learners to compare with predictions: 'Whose sketch matched? What would you change?'		glucose peaked at 15 min, confusing the units (minutes versus mmol/L), or saying the graph shows how much glucose was eaten. Use quick thumbs-up or thumbs-down checks after each calculation.
Explain Phase  VIDEO: Worked example: slope from graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this</i>	In groups of four, learners construct a first-draft explanation of the graph using a Claim-Evidence-Reasoning organiser. Claim: what causes the rise and what causes the fall? Evidence: specific numbers from the table and facts from the meal card. Reasoning: 'I think ... because ...'. They also respond to three prompts: (1) Which chemicals in the meal could be the source of the glucose? (2) What might the body be doing between 0 and 60 min? (3) What might be happening between 60 and 180 min so that glucose does not stay at 8.0	Say: 'Your job is not to be right today. Your job is to use evidence from the graph and the meal to give your best explanation so far.' Provide the CER organiser and sentence starters: 'The data show ...', 'This suggests ...', 'I am not sure about ...'. Circulate and ask probing questions: 'You said ugali causes the rise. What evidence from the meal card supports that?' 'You said the body removes glucose. Where might it go?' 'If glucose only came from the food, why would it ever come down?' After 8 minutes, WAIT TIME of 10-15 seconds, then cold-	Claim-Evidence-Reasoning (CER) with a confidence sort. Writing a claim and linking it to numbers pushes learners from vague talk to evidence-based reasoning. Marking sure and unsure statements makes gaps in knowledge visible, which sets up the questions for the DQB and shows learners that not knowing is the starting point of science.	Look for: (1) claims that identify food (especially ugali and other carbohydrates) as a source of glucose; (2) evidence that cites at least one number from the table; (3) acknowledgement that something must remove glucose from the blood after the peak; (4) a clear 'not sure' list that includes the processes (how food becomes glucose, where glucose goes). Mark each group on a 3-point scale: 1 = claim only, 2 = claim plus a number, 3 = claim plus number plus reasoning. Note learners who think glucose just 'disappears' or is 'burnt off' in the

activity. Search ARES for readings Search terms: chemicals life blood glucose	mmol/L? Each group marks statements they are confident about in one colour and statements they are unsure about in another. One member from each group shares the explanation in 1 minute. Learners then write in their notebooks 'What I am sure about' and 'What I am not sure about yet'.	call 4 groups to share. Record key ideas on chart paper under 'Our first ideas', writing exactly what learners say, even if partly incorrect. Ask: 'Which ideas did more than one group share?' and 'Which ideas disagree?' Do not give the scientific explanation. Say: 'These disagreements are exactly what we will test in the next lessons.' Praise the use of evidence, for example: 'I like how you used 3.5 mmol/L in 60 minutes versus 120 minutes.'		stomach.
Driving Question Board (DQB) Creation  VIDEO: Reading line graphs Source: Khan Academy (English - US curriculum) Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: chemicals life blood glucose	Each learner writes 2 or more questions on separate sticky notes, beginning with 'What', 'Why', 'How' or 'Which', that could help explain the glucose graph (for example: What chemicals are in ugali? Why does glucose rise slowly after eating? How does food become glucose? What makes the glucose go down? Which foods make glucose rise most?). Learners post their questions under the driving question: 'What chemicals are in the food we eat, and what happens to them in our bodies?' The class works together to group similar questions into categories (for example: What chemicals are in food? How are they broken down? Where does glucose go? What controls the level?). Learners vote with a star on the 3 questions they most want answered. They then mark which questions might be answered by testing foods in the lab and which need other kinds of evidence.	Display the driving question on a large chart. Say: 'Write every question you have. There are no silly questions.' Model one good question: 'What chemicals are in ugali?' and one that is too vague: 'Why is food?' and ask 'How could we make this one more testable?' Give 5 minutes of silent writing, then ask learners to post questions. Cold-call 5 learners to read their question aloud and say where it should go. Ask: 'Which questions are asking the same thing? Who can group them?' WAIT TIME of 10-15 seconds before each cold-call. Add your own question if missing: 'Why does the level not stay at 8.0 mmol/L?' Say: 'This board belongs to all of us. We will add answers and new questions in every lesson.' Link the questions to the KICD key inquiry questions: 'What are the chemicals of life? How do the chemicals of life contribute to the functioning of living organisms?' Photograph or keep the board safely for the next lessons. Tell learners that Lesson 2 will start from the question 'What chemicals are in food?'	Driving Question Board with question categorising. Learner-generated questions give ownership of the learning and show how the unit's lessons will answer them. Sorting questions into categories helps learners see the structure of the problem, namely chemicals in food, digestion, glucose in blood and control of glucose, and turns curiosity into an investigation plan.	Check that: (1) each learner has posted at least 2 questions; (2) at least half of the questions are testable or investigable rather than yes/no or factual lookups; (3) the board includes questions across at least 3 categories (food chemicals, breakdown, control of glucose); (4) learners can justify the category for their question. Record the number of learners who ask about what happens inside the body after the peak, which shows they are connecting to the fall in the graph. Give a quick prompt to learners with weak questions: 'Can you start it with How or Why?'
Model	Each learner draws an initial	Draw the skeleton on the board:	Developing and using models	Check that each model: (1) has

Building Phase  VIDEO: Reading line graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: chemicals life blood glucose	model on a half page of paper or in their notebook. The model is a flow diagram with boxes and arrows: Food chemicals (ugali, beans, sukuma wiki, oil) → ? → Glucose in blood (rises to 8.0 mmol/L at 60 min) → ? → Back to normal (4.5 mmol/L at 180 min). Learners put question marks where they do not know the process and add labels such as 'stomach?', 'cells?' or 'liver?' if they have ideas. In pairs they compare models and add one idea or question from their partner. Then each learner completes the Summary Table in their notebook with three columns: We observed, We learned, We can explain (see prompts). The class keeps the models in a folder to revise in Lesson 2 and later. Final celebration: learners stand and say together, 'We didn't just learn ABOUT science, we DID science!'	Food chemicals → ? → Glucose in blood → ? → Back to normal. Say: 'A model is our best current idea of how something works. It will change as we find evidence.' Remind learners to include numbers from the graph (4.5, 8.0 at 60 min, 4.5 at 180 min). Circulate and ask: 'What do you think happens at the first arrow? What goes in the second question mark?' Do not supply answers; if learners ask, say 'Good question, put it on the DQB.' After 8 minutes, WAIT TIME of 10-15 seconds, then cold-call 3 learners to describe their model in 30 seconds each. Ask: 'What is one thing you are unsure about in your model?' Collect or photograph models for comparison later. Close with the celebration checklist: 'Do you feel accomplished? Do you value the process? Are you confident to keep investigating?' and the closing statement: 'You didn't just learn ABOUT science, you DID science!' Preview Lesson 2: 'Next time we find out what the meal itself is made of.'	(initial flow-diagram model). Putting question marks in the model makes learners' current thinking and knowledge gaps visible. Because the same model is revised in every lesson, learners can see their own learning grow, and the model links the three pieces of the phenomenon: food chemicals, the rise in glucose, and the fall back to normal.	the five parts in correct order, from food chemicals to normal; (2) includes at least one number from the graph; (3) shows at least one question mark or unknown; (4) includes some idea about what causes the fall (for example cells, body, organ). Use a 3-level scale: Level 1 = boxes only, Level 2 = boxes with labelled processes, Level 3 = labelled processes plus evidence from the graph. Check the Summary Table for accurate data statements (peak 8.0 mmol/L at 60 min; rise 3.5 mmol/L; fall over 120 min). Note learners who confuse the rise with the fall or time with concentration.
---	--	--	--	--

D. TEACHER REFLECTION

After the lesson, consider: (1) Did every learner commit to a prediction before seeing the data, and how many predicted up then down? (2) Could learners calculate the rise (3.5 mmol/L) and the fall (3.5 mmol/L over 120 min) and say that the rise is faster than the fall? Which learners need support with percentage change (about 78%)? (3) What ideas did learners offer about the cause of the rise and fall (for example food, stomach, sugar, energy, 'burnt')? Record these to compare with evidence in later lessons. (4) Were the DQB questions testable, and do they cover chemicals in food, breakdown, where glucose goes, and what controls it? Which questions need rewording? (5) Were quieter learners heard through cold-calls and pair talk? (6) Were any learners uncomfortable discussing diabetes or fasting, and how did you handle it? (7) Is the DQB displayed safely so it can be revisited at the start of Lesson 2, and are the models stored for revision?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that Wanjiru's blood glucose concentration was 4.5 mmol/L before the meal of ugali, sukuma wiki and beans, rose to a peak of 8.0 mmol/L at 60 minutes (a rise of 3.5 mmol/L, about 78% above the start), then fell back to 4.5 mmol/L by 180 minutes
-----------------------------------	---

	(a fall of 3.5 mmol/L over 120 minutes). The rise was faster than the fall, and the level was back in the normal range (4.0 to 5.5 mmol/L) by 150 minutes at 4.7 mmol/L.
What did I learn?	We learned how to read a blood glucose graph and table, to calculate the change in concentration and compare how fast it rose and fell, and to turn what we wonder into testable questions. We learned that the meal contains different chemicals (about 70 g carbohydrate in ugali, about 20 g protein and 15 g carbohydrate in beans, about 10 g fat from oil, and vitamins and mineral salts in sukuma wiki) and that we do not yet know which ones affect blood glucose.
How does this explain the phenomenon?	We can explain, as a first idea, that something in the meal (probably the food chemicals in ugali) causes glucose in the blood to rise, and that something in the body must later lower it again because the level does not stay at 8.0 mmol/L. We cannot yet explain how food becomes glucose or where glucose goes, so we recorded these as questions on the Driving Question Board and in our initial flow-diagram model.

LESSON 2 (40 minutes (one double period may be used if available)): Elements and Compounds of Life: What Is Wanjiru's Lunch Made Of?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners identify the elements that make up living things and foods, and classify the nutrients in Kenyan foods (ugali, beans, sukuma wiki, milk, nyama choma) as organic or inorganic compounds, so that they can say which chemicals might be responsible for the rise and fall of Wanjiru's blood glucose.
Knowledge	Living things and foods are made of elements: carbon (C), hydrogen (H), oxygen (O), nitrogen (N), phosphorus (P), sulphur (S), plus mineral elements Ca, Fe, Mg, Na and K. Elements combine to form compounds. Organic compounds contain carbon bonded to hydrogen: carbohydrates, proteins, lipids and vitamins. Inorganic compounds include water and mineral salts. The meal of ugali 300 g, sukuma wiki 100 g and beans 100 g contains about 70 g carbohydrate in ugali, about 20 g protein and 15 g carbohydrate in beans, about 3 g protein plus fibre, vitamins A and C and mineral salts (iron, calcium) in sukuma wiki, and about 10 g fat from cooking oil.
Skills	Developing and using models (a classification chart of food chemicals); analysing and interpreting data (meal nutrient approximations); constructing explanations; classifying; asking questions; communicating scientific ideas; collaboration.
Attitudes	Curiosity about what everyday foods are made of; respect for others' ideas during group classification; appreciation of Kenyan foods as sources of the chemicals of life; responsibility for evidence-based claims.
Key Inquiry Question	What are the chemicals of life? (KICD key inquiry question 1). Lesson question: Which chemicals are in Wanjiru's lunch, and which are organic and which are inorganic?
Purpose in Storyline	In Lesson 1 learners read Wanjiru's blood glucose graph (4.5 mmol/L at 0 min, a peak of 8.0 at 60 min, back to 4.5 by 180 min) and the DQB questions asked what chemicals are in food. This lesson answers the first part: it names the elements and compound groups in the meal and sorts them as organic or inorganic. It narrows the list of suspects for the glucose spike to organic compounds, especially carbohydrates, but does not yet test for them (Lesson 3) or explain proteins and lipids in detail (Lesson 4) or enzymes (Lesson 5).
Safety Notes	No chemical reagents are used in this lesson. Learners handle only food samples for display: ugali, cooked beans, sukuma wiki, milk, nyama choma. Use clean plates and hand washing before and after. Do not allow tasting unless the food is hygienically prepared and no learner has an allergy (check for groundnut, milk or other allergies). Keep samples covered and dispose of leftovers responsibly.

B. LESSON OVERVIEW

Mr Otieno brings Wanjiru's lunch to class as food samples (or labelled pictures): ugali, beans, sukuma wiki, milk and nyama choma. Learners first predict what chemicals the foods contain, then collect evidence from the meal nutrient approximations and a short element card activity showing that all foods are made of a small set of elements (C, H, O, N, P, S, Ca, Fe, Mg, Na, K). In the explain phase learners build a classification chart sorting nutrients into organic (carbohydrates, proteins, lipids, vitamins) and inorganic (water, mineral salts) groups, using the rule that organic compounds contain carbon bonded to hydrogen. They return to the DQB to answer 'What chemicals are in food?' at the level of compound groups and add new questions about how to detect each group, and add the first box of the class model: Food chemicals.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Elemental building blocks of biological molecules Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Elements of life Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners see Wanjiru's lunch on a table: ugali, cooked beans, sukuma wiki, milk and nyama choma (real samples or pictures). Mr Otieno reminds them of the graph from Lesson 1, where blood glucose rose from 4.5 to 8.0 mmol/L at 60 min. Individually, in their science notebooks, learners write for 3 minutes: 'List the chemicals you think are in each food.' Then they predict: 'Which of these chemicals might make blood glucose rise?' They draw a quick two-column table, with foods on the left and predicted chemicals on the right (for example: sugar, starch, protein, fat, water, salt, iron). In pairs they compare lists and circle one chemical they are unsure about.	Display the foods and the Lesson 1 graph. Say: 'Last lesson we asked what chemicals are in the food Wanjiru ate. Today we start answering. Before I tell you anything, write what you already think.' Give 3 minutes of silent writing. Then say: 'Turn to your partner and compare. Where do you agree? Where do you disagree?' Cold-call 4 learners (not volunteers) to share one chemical each; record all ideas on the board without judging, including incorrect ones such as 'sugar is in ugali'. Ask: 'What makes you think that?' and WAIT TIME 10-15 seconds before taking responses. Ask: 'Which of these chemicals could end up as glucose in the blood?' and record 3 predictions. Circulate and note common ideas, such as 'chemicals are only in medicines' or 'ugali is just one chemical'.	Eliciting prior knowledge through a Predict-and-Record (individual then pair then whole class). Writing predictions first makes learners commit to ideas, so they can later see how evidence changes their thinking. Recording all ideas, including misconceptions, creates a visible baseline for revision of the class model and the DQB.	Look for: (1) at least 4 of 5 foods with a predicted chemical in each learner's table; (2) use of words such as starch, sugar, protein, fat, water, minerals; (3) a reason given for the glucose prediction (for example 'ugali is starchy'). Note how many learners name more than one chemical per food and how many think a food is only one chemical. Tally misconceptions on the board to revisit at the end of the lesson.
Observe Phase  VIDEO: Elemental building blocks of biological molecules Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Biochemical Compound Classification (Flexbook) Source: CK-12  Search ARES for similar readings	Learners work in groups of 4 with three evidence sources. Station A, Nutrient data: a card with the meal nutrient approximations: ugali about 70 g carbohydrate (mostly starch from maize); beans about 20 g protein and 15 g carbohydrate; sukuma wiki about 3 g protein plus fibre, vitamins A and C and mineral salts (iron, calcium); cooking oil about 10 g fat. Groups also record milk and nyama choma as sources of protein, fat and minerals (calcium in milk, iron in nyama choma) from the supplied food label cards. Station B, Element cards: learners receive cards for C, H, O, N, P, S, Ca, Fe, Mg, Na, K with a short note on where each is found (for example C, H, O in starch and glucose; N	Introduce: 'Wanjiru's lunch is not one chemical. It is many. You are chemical detectives today.' Distribute cards and set a 15 minute rotation (5 min per station). At Station A ask: 'Which food has the most carbohydrate? How many grams? How does that compare with the protein and the fat?' Use cold-calls: after the rotation, cold-call 3 groups to report one fact from Station A and 2 groups for Station B. At Station C do the burn demonstration yourself; keep learners 2 metres away and keep long hair tied back. Ask: 'What is the black material? What does it tell us about what the food contains?' and WAIT TIME 10-15 seconds. Prompt: 'Which element is in every organic nutrient on your	Data and Evidence Gathering with a Jigsaw Rotation (Analysing and Interpreting Data). Learners gather evidence from numbers, element information and a demonstration. Linking charring to carbon gives physical evidence for the idea that organic compounds contain carbon, rather than being told. Quantity data (70 g, 20 g, 15 g, 10 g, 3 g) lets learners compare the amounts of each nutrient in the meal and begin to see which may matter most for the glucose spike.	Check observation tables for: (1) correct amounts copied from the meal data (ugali about 70 g carbohydrate, beans about 20 g protein and 15 g carbohydrate, about 10 g fat); (2) at least 3 correct element-food matches; (3) the statement that black residue shows carbon in the food. Listen for: 'ugali has the most carbohydrate' and 'salt did not turn black'. If a group cannot say which element is in all organic compounds, ask them to re-read the element cards and the burn test results.

	and S in protein; Ca in milk; Fe in sukuma wiki and nyama choma). They match each element to at least one food in the meal. Station C, Burn test demonstration: Mr Otieno holds a small piece of dry maize flour (or ugali crust) on a spatula in a flame at a safe distance (teacher only), and learners observe blackening (carbon) and discuss what the black residue shows, and compare with table salt, which does not blacken. Groups record observations in a table and answer: 'Which foods contain the most carbohydrate? Which element is found in all the organic nutrients?'	card?' Correct gently if learners say salt is organic: 'Does salt blacken? What does that suggest about carbon?' Collect each group's observation table.		
Explain Phase  VIDEO: Photosynthesis <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Biochemical Compound Classification Practice Source: CK-12  Search ARES for similar readings	Using their evidence, groups build a large Classification Chart on sugar paper with two columns, Organic and Inorganic. They write the nutrient groups found in the meal: carbohydrates (starch in ugali, carbohydrate in beans), proteins (beans, milk, nyama choma), lipids (cooking oil, milk fat), vitamins (A and C in sukuma wiki), water (in all foods, especially milk and sukuma wiki), and mineral salts (iron, calcium in sukuma wiki; calcium in milk). They write the rule: 'Organic compounds contain carbon bonded to hydrogen; inorganic compounds such as water and mineral salts do not.' Next to each group they list the key elements (for example carbohydrates: C, H, O; proteins: C, H, O, N, sometimes S; water: H, O). Then each learner writes a Claim-Evidence-Reasoning statement: 'Claim: The chemical in Wanjiru's meal most likely to cause the glucose rise is _____. Evidence: _____. Reasoning:	Say: 'Use only your evidence from today to decide where each nutrient goes. Be ready to defend your choice.' Circulate and ask: 'Which element is the clue that something is organic?' and 'Why is water placed in the inorganic column even though it is found in every living thing?' Give WAIT TIME of 10-15 seconds after each question. Hold a gallery walk of 5 minutes and cold-call 4 learners to explain a classification choice from another group. Clarify: 'Carbon alone is not enough; organic compounds have carbon bonded to hydrogen.' Address the misconception that organic means grown without chemicals: 'In Biology, organic refers to compounds that contain carbon bonded to hydrogen.' For CER, say: 'Your evidence must come from the data or the burn test.' Do NOT test starch or sugar yet; if learners ask how we know ugali has starch, reply: 'Great question. We will test that in the next lesson.	Modelling (classification chart) combined with Claim-Evidence-Reasoning (CER). The chart turns scattered evidence into a model showing how elements build compounds and how compounds group as organic and inorganic. CER forces learners to link the glucose spike to a specific chemical group using the 70 g carbohydrate figure, building scientific argumentation. Peer critique with a checklist exposes weak reasoning (for example claiming protein is the cause without evidence).	Success indicators: (1) chart places carbohydrates, proteins, lipids and vitamins under Organic and water and mineral salts under Inorganic with no more than 1 error; (2) the rule 'carbon bonded to hydrogen' is written correctly; (3) CER names carbohydrate (or starch in ugali) as the most likely cause, citing 70 g in ugali; (4) elements are written with correct symbols (C, H, O, N, P, S, Ca, Fe, Mg, Na, K). If learners place vitamins or mineral salts incorrectly, use a quick exit question: 'Does vitamin C contain carbon? Does iron salt?'

	____.' Pairs critique each other's statements using a checklist.	Add it to the DQB.' Do not teach protein or lipid structure; say that this comes later in the sequence.		
Driving Question Board (DQB) Creation  VIDEO: Biogeochemical cycles overview <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Elements of life Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the class DQB started in Lesson 1. In their groups they take 2 sticky notes of one colour and write answers or partial answers to earlier questions such as 'What chemicals are in food?' (for example: 'carbohydrates, proteins, lipids, vitamins, water, mineral salts'). With a second colour they write 2 new questions that came up, such as 'How do we know ugali has starch?', 'Is starch the same as glucose?', 'What do proteins and fats do in the body?', 'How does the body break these chemicals down?' Groups place notes on the board and the class sorts the new questions into categories: Testing for chemicals, What chemicals do, and What happens to chemicals in the body. A volunteer marks questions that this lesson helped answer with a tick.	Say: 'Go back to the questions we asked about Wanjiru's graph. Which ones can we now partly answer? Which new questions do we have?' Give 4 minutes for writing. Cold-call 3 groups to read one question aloud and say how it links to the graph. WAIT TIME 10-15 seconds before asking: 'What do we still not know about why glucose rose and then fell?' Help learners sharpen vague questions: 'You wrote What is in food? Can you make it testable, for example How can we tell if ugali contains starch?' Group the questions and point out that the testing questions will guide Lesson 3 (starch and sugar tests) and the others will be answered in later lessons. Do not answer them now.	Question tracking with the DQB. Returning to the board connects each piece of evidence to the bigger puzzle of Wanjiru's graph and shows learners how their understanding is growing. Turning gaps into testable questions develops the practice of asking questions and defining problems, and gives a visible purpose for the next lesson.	Observable indicators: (1) every group posts at least 1 answer note using today's vocabulary (organic, inorganic, carbohydrate, protein, lipid, mineral salts); (2) at least 2 new questions per group are testable or link to the graph; (3) learners can explain which DQB questions are partly answered. Note groups whose questions remain vague (for example 'Why is food important?') and coach them to rewrite them.
Model Building Phase  VIDEO: Cellular respiration <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Biochemical Compound Classification (Flexbook)	Learners open the class model, a flow diagram: Food chemicals, then digestion by enzymes, then glucose in blood, then cells (respiration) and storage, then back to normal. In this lesson they fill in only the first box, Food chemicals. Groups add to a large class diagram: 'Food chemicals: organic (carbohydrates, proteins, lipids, vitamins) and inorganic (water, mineral salts) from ugali, beans, sukuma wiki, milk and nyama choma.' They add the amount labels (ugali about 70 g carbohydrate; beans about 20 g protein and 15 g carbohydrate; about 10 g fat) and mark carbohydrates with a star as 'most	Say: 'Our model is a first draft. Today we can only fill in the first box. What evidence supports each thing we write?' Ask groups to justify each addition. Cold-call 3 learners: 'What did you add to the model and what evidence supports it?' WAIT TIME 10-15 seconds. Ask: 'Why have we not filled in the box for glucose in blood yet? What do we still need to find out?' Return to the Predict Phase list and ask 2 learners to name one idea they have revised. Stress: 'We know what kinds of chemicals are in Wanjiru's lunch, but we do not yet know which one turns into glucose or how.' Close the lesson with the celebration: 'You didn't	Cumulative model revision (flow diagram) with a Summary Table. Adding only the evidence-supported first box teaches learners that models grow with evidence and have honest gaps. Comparing the model with their initial predictions makes learning visible and supports metacognition. Leaving later boxes as question marks creates a need to know for Lessons 3 to 6.	Look for: (1) the Food chemicals box correctly lists organic and inorganic groups; (2) amounts match the meal data; (3) learners write a revised idea (for example 'I thought ugali was one chemical but it contains starch, a carbohydrate'); (4) learners identify at least one gap in the model. Exit slip: 'Name two organic and two inorganic chemicals in Wanjiru's meal and the element that all organic compounds contain.' Success: at least 80% correct (carbohydrate, protein, lipid or vitamin as organic; water and mineral salts as inorganic; carbon).

Source: CK-12 Search ARES for similar readings	likely linked to the glucose rise'. Boxes for digestion, glucose in blood and cells stay as question marks. Each learner copies the model in their notebook and writes one sentence on what changed in their thinking since the Predict Phase. Finally, the class completes the Summary Table (What we observed, What we learned, How we can explain it).	just learn ABOUT science—you DID science!' and ask learners to check the boxes: I feel accomplished, I value the process, I am confident.		
---	---	---	--	--

D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners use evidence (nutrient amounts, burn test) rather than guesses to classify nutrients? (2) Which misconceptions remained, for example that 'organic' means natural or that water is organic? (3) Did the burn demonstration help link carbon to organic compounds? (4) Were all learners participating in cold-calls and group tasks, including quieter learners? (5) Are the DQB questions testable enough to drive Lesson 3? (6) Did anyone confuse carbohydrate with glucose in the model, and how will you address it in Lesson 3? Note adjustments to timing, grouping and food samples, considering allergies and available local foods.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe? Record the amounts of each nutrient in Wanjiru's meal (ugali about 70 g carbohydrate; beans about 20 g protein and 15 g carbohydrate; sukuma wiki about 3 g protein plus fibre, vitamins A and C, iron and calcium; about 10 g fat from cooking oil) and what happened when dry maize flour was burnt (it blackened, showing carbon).
What did I learn?	What did we learn? Foods and living things are made of elements (C, H, O, N, P, S, Ca, Fe, Mg, Na, K) that form compounds. Organic compounds (carbohydrates, proteins, lipids, vitamins) contain carbon bonded to hydrogen; inorganic compounds (water, mineral salts) do not.
How does this explain the phenomenon?	How can we explain Wanjiru's glucose graph so far? Her lunch contains several organic and inorganic chemicals, and the largest amount is carbohydrate (about 70 g in ugali), so carbohydrate is the most likely cause of the rise from 4.5 to 8.0 mmol/L. We still need tests and more evidence to show how it becomes glucose and why the level falls back.

LESSON 3 (80 minutes (double lesson)): Carbohydrates: Starch and Sugars

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners test foods for starch and reducing sugars, then use their evidence to link the starch in ugali to the rise in Wanjiru's blood glucose concentration from 4.5 to 8.0 mmol/L.
Knowledge	Carbohydrates are organic compounds made of carbon, hydrogen and oxygen. Starch is a large carbohydrate made of many glucose units (monosaccharides) joined together. Iodine turns blue-black when starch is present. Benedict's solution changes from blue through green, yellow and orange to brick-red as the amount of reducing sugar increases. Ugali contains about 70 g of carbohydrate, mostly starch from maize.
Skills	Planning and carrying out an investigation (NGSS SEP 3); safe use of a water bath, test tubes and reagents; recording observations in a results table; analysing data and interpreting a graph (SEP 4); constructing explanations from evidence (SEP 6); developing and revising a model (SEP 2).
Attitudes	Curiosity about what happens to the food we eat; care and responsibility when handling chemicals and hot water; respect for the ideas of peers; appreciation of evidence as the basis of scientific claims.
Key Inquiry Question	What are the chemicals of life? How do the chemicals of life contribute to the functioning of living organisms?
Purpose in Storyline	Lesson 1 gave learners the blood glucose graph (spike to 8.0 mmol/L at 60 min, then fall to 4.5 at 180 min) and the question of which food chemicals cause it. Lesson 2 showed that foods contain organic compounds, including carbohydrates. In this lesson learners gather the first direct chemical evidence: ugali and maize flour contain starch, and starch is made of glucose units. This links ugali to the glucose spike. Lesson 4 will turn to proteins and lipids, and Lesson 5 to how enzymes digest starch.
Safety Notes	Wear safety goggles and aprons. Iodine solution stains skin and clothes and must not be tasted. Benedict's solution is heated only in a water bath (about 80 degrees C), never directly in a flame. Point test tubes away from faces. Use test-tube holders. Do not eat any test food in the laboratory. Wipe up spills at once and wash hands after the practical. Dispose of the used reagents as directed by Mr Otieno.

B. LESSON OVERVIEW

Mr Otieno displays Wanjiru's blood glucose graph again and asks learners what in her lunch of ugali, sukuma wiki and beans could have produced the rise of 3.5 mmol/L in 60 minutes. Learners predict which test foods contain starch and reducing sugar. They then carry out the iodine test and Benedict's test on ugali, maize flour, sukuma wiki, ripe banana, milk, nyama choma and a glucose solution. Next they use their results table and a glucose-chain model to explain how starch in ugali (about 70 g carbohydrate in the meal) is related to glucose in the blood. Important: ugali gives no change with Benedict's before digestion, so learners must reason about what must happen to starch before glucose can enter the blood. Learners add new questions to the DQB and revise the class flow-diagram model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners look again at Wanjiru's graph (0 min: 4.5; 60 min: 8.0)	Display the graph and say: "Wanjiru's blood glucose rose by	Prediction with reasoning (Think-Write-Pair-Share): committing to a	Check that 100% of learners have written predictions with a reason.

 VIDEO: Molecular structure of glucose Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Carbohydrates Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	peak; 180 min: 4.5 mmol/L). Individually, in 3 minutes, they write: (1) which of these foods they think contain starch: ugali, maize flour, sukuma wiki, ripe banana, milk, nyama choma; (2) which contain reducing sugar (such as glucose); (3) what colour they expect iodine to turn if starch is present. They record predictions in a table with a column for their reasoning. They then share with an elbow partner and mark each prediction with a confidence level (sure, think so, guess).	3.5 mmol/L in 60 minutes after her lunch. Which food on her plate do you think is mostly responsible? Write your prediction and your reason before we talk." Give 3 minutes of silent writing, then WAIT TIME of 10-15 seconds after asking, "Which foods do you predict contain starch?" Cold-call 4 learners (not volunteers) and record every prediction on the board without correcting. Ask: "What makes you say that?" Remind learners of Lesson 2: carbohydrates are organic compounds found in food. Say: "Today we will test, not guess. Our predictions are our hypotheses." Circulate and note learners who link ugali to maize or who think only sweet foods contain carbohydrate.	prediction and a reason exposes learners' existing ideas, such as that only sweet foods are carbohydrates, so the later evidence can confirm or challenge them. It makes the investigation purposeful because learners want to know whether they were right.	Look for: (a) learners who predict ugali or maize flour contain starch; (b) learners who connect a food to the glucose rise; (c) learners who think ugali contains reducing sugar (a misconception to revisit). Tally predictions on the board; at least 3 of 4 cold-called learners should give a reason, not only a food name.
Observe Phase  VIDEO: Introduction to carbohydrates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Carbohydrate Classification (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of 4, learners carry out two food tests. Test 1, iodine test for starch: place a small amount of ugali, maize flour, sukuma wiki leaf (crushed), milk and nyama choma on a white tile or in separate test tubes with water, add 2-3 drops of iodine solution and record the colour (blue-black = starch present). Test 2, Benedict's test for reducing sugars: put 2 cm³ of each food solution (ripe banana mashed in water, ugali, milk, glucose solution) into test tubes, add 2 cm³ of Benedict's solution, heat in a hot water bath for 3-5 minutes and record the colour (blue to green to yellow to orange to brick-red = increasing reducing sugar). Expected results: iodine: ugali blue-black, maize flour blue-black, sukuma wiki slight or negative, milk negative, nyama choma negative. Benedict's: ripe banana brick-red/orange, ugali	Model safe handling first: "Goggles on. Iodine stains. Heat only in the water bath, never in the flame." Assign roles: materials manager, recorder, timer/safety officer, reporter. While groups work, ask: "What colour do you see? How would you describe it to someone who cannot see it?" and "What did you expect for ugali? Does your result match?" For groups whose ugali gives no change with Benedict's, say: "Interesting. Ugali is full of carbohydrate, yet no brick-red. What might that tell us?" Do not give the answer. WAIT TIME of 10-15 seconds before taking responses. After the practical, cold-call 3 groups to report one result each and write the results on a class table. Ask the class to compare: "Which foods gave a positive iodine test? Which gave a positive Benedict's test? Did any food give a positive result for both	Collaborative data collection and class data table (Gallery of results): pooling every group's observations lets learners see patterns across foods, such as starchy foods giving blue-black with iodine, and notice the puzzle that ugali is positive for starch but negative for reducing sugar. This conflict drives the next phase and builds understanding of evidence-based claims.	Observe whether learners (a) use the iodine and Benedict's tests correctly, with correct volumes in cm³ and the water bath; (b) record colours accurately using the correct colour scale; (c) follow safety rules. Collect each group's results table: at least 80% of entries should match expected results (ugali blue-black; banana brick-red/orange; glucose brick-red). Note groups that can state which test is for starch and which is for reducing sugar.

	blue (no change), milk green or yellow (lactose, weak positive), glucose brick-red. Learners record observations in a results table and compare with their predictions.	or neither?" Introduce the term 'reducing sugar' for a Benedict's positive result and 'starch' for a positive iodine result.		
Explain Phase  VIDEO: Molecular structure of glucose Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Carbohydrate Classification Practice Source: CK-12  Search ARES for similar readings	Learners use the class results table to explain the link between ugali and the glucose spike. Step 1: they answer in groups: "Ugali is blue-black with iodine but blue with Benedict's. What does this tell us about the type of carbohydrate in ugali?" (starch present, but no reducing sugar before digestion). Step 2: using a paper-strip or bead model, learners link 8-10 'glucose' beads or paper units into a chain to model starch and then separate them to model digestion into glucose, noting that glucose is a monosaccharide and gives a brick-red result. Step 3: they use the numbers: ugali has about 70 g carbohydrate, mostly starch from maize; Wanjiru's blood glucose rose by 3.5 mmol/L (from 4.5 to 8.0) in 60 minutes. Each learner writes a claim-evidence-reasoning (CER) statement: Claim: starch in ugali is the main source of the glucose that raised the blood glucose concentration. Evidence: from the tests and the data. Reasoning: starch is made of glucose units; it must be broken down to glucose before it enters the blood. Learners also name the carbohydrates seen today: starch (ugali, maize flour), glucose (monosaccharide), lactose in milk (weak positive) and sugar in ripe banana.	Pose: "Ugali has about 70 g of carbohydrate but showed no reducing sugar. Yet Wanjiru's blood glucose went up. What must have happened to the starch?" Give WAIT TIME of 10-15 seconds, then ask learners to turn and talk for 2 minutes. Cold-call 4 learners to share and probe: "What evidence from our tests supports that?" Distribute the bead or paper chains and say: "Each bead is one glucose unit. Starch is many joined together. What would you need to do to get single glucose units?" Link to the data: "The rise was 3.5 mmol/L in 60 minutes. Does the meal contain enough carbohydrate to explain it?" Remind learners that beans also have about 15 g carbohydrate, so ugali is the main but not the only source. Do NOT teach how enzymes digest starch or what insulin does; say: "We will find out how in later lessons. For now, write it as a question on the DQB." Use sentence frames for CER: "I claim... because the test showed... This means..." Use the exemplar 'Mama Achieng sells maize near Kisumu, and the maize flour test showed it is rich in starch' as a local application.	Claim-Evidence-Reasoning with physical modelling: the bead chain makes the abstract idea that large starch molecules are made of many glucose units concrete, and the CER structure forces learners to connect test colours (evidence) to the glucose spike (claim) with scientific reasoning, including the discrepancy that starch is not itself a reducing sugar.	Look for CER statements that include (a) a claim naming starch or ugali; (b) evidence citing the blue-black iodine result and the no-change Benedict's result for ugali, or the 70 g carbohydrate; (c) reasoning stating that starch is made of glucose units that must be broken down. Success indicator: at least 75% of learners write a complete CER, and cold-called learners correctly say glucose is a reducing sugar and starch is not. Note any learners who say ugali contains glucose already; revisit with the Benedict's evidence.
Driving Question Board (DQB) Creation	Learners revisit the class DQB. On sticky notes they (1) write answers they can now give with evidence (for example, 'Ugali and maize	Say: "Look at our DQB. Which questions can we now answer with evidence? Move those notes to the 'Answered' column." Give WAIT	Driving Question Board as a public record of growing understanding: moving questions from 'unanswered' to 'answered' with	Check that each learner contributes at least one note and that at least 2 new questions are about the mechanism (how starch

 VIDEO: Introduction to carbohydrates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Carbohydrates (Flexbook) Source: CK-12  Search ARES for similar readings	flour contain starch, shown by the blue-black iodine result'; 'Glucose gives a brick-red Benedict's result', and (2) write new questions that arose, such as: 'How does starch become glucose?', 'What breaks starch into glucose in our bodies?', 'Does the body use only carbohydrates for energy?', 'Why did the glucose fall after 60 minutes?', 'Why did protein foods like beans and nyama choma not test positive for starch?' The class sorts the notes into groups: answered, partly answered and still to investigate, and links each to a future lesson.	TIME of 10-15 seconds. Cold-call 3 learners to read an answered question and justify it with test results. Ask: "What new questions has today's evidence made you ask?" and collect at least 5 new notes. Group the questions into themes: how starch becomes glucose (Lesson 5), other food chemicals (Lesson 4) and what happens to glucose after absorption (Lesson 6). Do not answer these yet. Ask: "Which of our unanswered questions would you test next, and how?" Acknowledge learners whose questions build on others.	evidence shows learners their own progress, while new questions give purpose and direction for the next lessons and keep learners curious about the phenomenon.	becomes glucose, what happens next). Look for evidence-based wording in answered notes ('the test showed...'). Count how many questions link to the fall in glucose after 60 minutes; if fewer than 2, prompt learners to look again at the graph.
Model Building Phase  VIDEO: Dehydration synthesis or a condensation reaction <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Carbohydrates Practice Source: CK-12  Search ARES for similar readings	Learners revise the class flow-diagram model: Food chemicals, then digestion by enzymes, then glucose in blood, then cells (respiration) and storage, then back to normal. In groups they add today's evidence to the first box: 'Food chemicals: carbohydrates, including starch from ugali and maize (about 70 g carbohydrate in ugali), made of glucose units; starch gives blue-black with iodine; glucose gives brick-red with Benedict's.' Between the first box and 'glucose in blood' they draw an arrow with a question mark labelled 'starch must be broken into glucose: how?' They annotate the graph at the rise (4.5 to 8.0 mmol/L, +3.5 in 60 min) with 'glucose from carbohydrate in the meal'. Each group states one thing they are sure of, one thing they are unsure of and one prediction for Lesson 4. Learners complete the summary table (What we observed, What we learned, How it helps explain the phenomenon).	Say: "Take your model from the last lesson. What can you now add to the 'food chemicals' box, using today's test results as evidence?" Provide markers in a different colour so revisions are visible. Circulate and ask: "Where on your model is the starch from the ugali? Where is the glucose in Wanjiru's blood?" and "What is still missing between those two boxes?" WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to present one revision and justify it with evidence. Close with the template celebration: "You didn't just learn ABOUT science, you DID science!" Check the Final Celebration boxes: students feel accomplished, students value the process, students are confident. Preview: "Next time we will test the proteins and lipids in beans, groundnuts and mandazi. Will they explain the spike too?"	Cumulative model revision (flow diagram): adding evidence in a different colour makes learning visible and helps learners see how today's data fills one part of the chain and exposes the gap (starch to glucose) that later lessons will fill. Constructing and revising a model is a core science practice (SEP 2).	Look for models that (a) include starch and glucose in the correct boxes; (b) mention the iodine and Benedict's evidence; (c) show the gap between starch and blood glucose as a question. Success criteria: at least 80% of groups add carbohydrate evidence with correct terms and each group names one uncertainty. Use the summary tables as an exit ticket: learners should state that ugali contains starch, which is made of glucose units.

D. TEACHER REFLECTION

Did learners correctly distinguish starch (iodine test) from reducing sugar (Benedict's test)? Which groups struggled with the Benedict's colour scale or water bath safety? Did learners notice and try to explain that ugali is rich in carbohydrate but showed no reducing sugar? How many learners wrote CER statements that connected the test results to the 3.5 mmol/L rise? What misconceptions (only sweet foods are carbohydrates; ugali already contains glucose) remain? Which DQB questions show learners are ready to explore enzymes and other nutrients? What would I change in grouping, timing or materials, especially if ripe bananas or maize flour were scarce?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Iodine turned blue-black with ugali and maize flour (starch present) but not with milk or nyama choma, and only slightly or not with sukuma wiki. Benedict's solution turned brick-red/orange with ripe banana and brick-red with glucose, green or yellow with milk, and stayed blue with ugali.
What did I learn?	Carbohydrates are organic compounds. Starch is a large carbohydrate made of many glucose units; glucose is a reducing sugar. Ugali has about 70 g of carbohydrate, mostly starch from maize. Ugali tested positive for starch but not for reducing sugar before digestion.
How does this explain the phenomenon?	The starch in ugali is a source of the glucose that raised Wanjiru's blood glucose concentration from 4.5 to 8.0 mmol/L (a 3.5 mmol/L rise in 60 minutes), because starch is made of glucose units. We still need to find out how starch is broken down into glucose and what happens to glucose after it reaches the blood.

LESSON 4 (80 minutes (2 x 40-minute periods)): Proteins and Lipids: Do They Cause the Glucose Spike?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate proteins and lipids in beans, groundnuts, milk, nyama choma and mandazi, and compare them with carbohydrates to decide whether they cause the glucose spike in Wanjiru's graph.
Knowledge	Proteins are organic compounds made of amino acids and are detected by the biuret test (blue to violet/purple). Lipids are organic compounds made of fatty acids and glycerol and are detected by the grease-spot test (translucent spot) and the emulsion test (cloudy white layer). Carbohydrates (Lesson 3) are made of glucose units and are tested with iodine and Benedict's. Wanjiru's meal has about 20 g protein and about 10 g fat, compared with about 70 g carbohydrate in ugali alone, and the meal data link carbohydrates, not proteins or lipids, to the glucose rise from 4.5 to 8.0 mmol/L.
Skills	Carrying out the biuret, grease-spot and emulsion tests safely; recording results in a table; comparing results with evidence from Lesson 3; using evidence to argue which food chemicals cause the glucose spike (NGSS SEP: Planning and Carrying Out Investigations; Analyzing and Interpreting Data; Engaging in Argument from Evidence).
Attitudes	Curiosity about what is in everyday foods; responsibility and safety when handling chemicals; respect for peers' ideas during discussion; appreciation of the roles of proteins and lipids in the body.
Key Inquiry Question	1. What are the chemicals of life? 2. How do the chemicals of life contribute to the functioning of living organisms?
Purpose in Storyline	Lesson 2 classified carbohydrates, proteins and lipids as organic compounds, and Lesson 3 linked ugali starch to the glucose spike using iodine and Benedict's tests. Lesson 4 completes the main food chemical classes by testing proteins and lipids, showing they are chemically different and do not cause the spike, which sets up Lesson 5 (enzymes) and Lesson 6 (absorption, respiration, insulin).
Safety Notes	Wear safety goggles and aprons. Biuret reagent (or dilute sodium hydroxide with dilute copper sulphate) is irritant: wear gloves, avoid skin and eye contact, and rinse spills with plenty of water. Ethanol for the emulsion test is flammable: keep away from flames and use only small amounts (2 cm ³). Never taste any test material in the laboratory. Check for groundnut or milk allergies before the lesson and let affected learners handle only sealed samples or act as recorders. Wash hands after the practical and dispose of waste as directed by Mr Otieno.

B. LESSON OVERVIEW

Mr Otieno's class returns to Wanjiru's graph, which showed blood glucose rising from 4.5 to 8.0 mmol/L at 60 minutes and falling back to 4.5 mmol/L by 180 minutes after ugali (300 g), sukuma wiki (100 g) and beans (100 g). Learners predict whether the protein in beans and the oil in the meal could also cause the spike. They then carry out the biuret test on beans, milk, nyama choma and ugali, and the grease-spot and emulsion tests on groundnuts, mandazi, milk and ugali. In Explain they compare the structure and roles of proteins and lipids (amino acids; fatty acids and glycerol) with carbohydrates, and use the meal data (about 20 g protein, 10 g fat, 70 g carbohydrate in ugali) to argue that carbohydrates are the main cause of the spike. The DQB and the class flow-diagram model are then revised.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
-------	--------------------	---------------	----------------------	-------------------------------

Predict Phase  VIDEO: Introduction to proteins and amino acids Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Proteins Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look again at Wanjiru's glucose graph (0 min: 4.5; 15 min: 5.5; 30 min: 7.0; 60 min: 8.0 peak; 90 min: 6.5; 120 min: 5.2; 150 min: 4.7; 180 min: 4.5 mmol/L) and the meal list. Individually they write a prediction: 'Beans contain about 20 g protein and the meal has about 10 g fat from cooking oil. Could these also cause the glucose spike? Which of milk, nyama choma, groundnuts, mandazi and ugali will contain protein, and which lipid?' They sketch a prediction table (food: protein? lipid?) with a reason, then share with a partner (3 minutes) before the class vote.	Display the graph and say: "Last lesson we linked the 70 g of carbohydrate in ugali to this spike. But Wanjiru's plate also had beans and oil. Do those cause the spike too? Write your prediction and one reason." Give 2 minutes silent writing. WAIT TIME: 10-15 seconds after asking, "Which foods do you think contain protein?" before taking answers. Cold-call 4 learners from different groups, asking "What made you think that?" Record predictions on the board in two columns (protein / lipid) without correcting. Ask: "Which test method from Lesson 3 could we adapt to check?" and take 2 responses.	Prediction with reasoning (commit-then-test): writing a prediction and a reason before any evidence exposes prior ideas, such as 'all food becomes sugar', and gives learners a stake in the results, so the later data confirm or challenge their thinking.	Observable indicators: (1) each learner has a written prediction with a reason; (2) at least 3 learners link a test to a food chemical (for example 'iodine is for starch so we need a different test'); (3) the teacher records common misconceptions, such as 'beans are sugary so they cause the spike' or 'oil turns to glucose immediately', for revisiting in Explain.
Observe Phase  VIDEO: Introduction to proteins and amino acids Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Proteins Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4 learners carry out two tests. Test A, biuret test for protein: put about 2 cm³ of each food sample or extract (bean paste mixed with water, milk, minced nyama choma, ugali paste) in separate test tubes, add 2 cm³ of dilute sodium hydroxide, then add 2-3 drops of dilute copper sulphate and shake. Test B, lipids: (i) grease-spot test: rub or crush groundnuts, mandazi, a drop of milk and ugali on separate pieces of brown or filter paper, dry gently, and hold up to the light; (ii) emulsion test: shake the sample with 2 cm³ of ethanol, then pour the clear liquid into 2 cm³ of cold water and look for a cloudy white layer. They record results in a table. Expected results: biuret: beans violet/purple, milk violet, nyama choma violet, ugali pale blue/faint (negative or weak). Grease spot: groundnuts translucent spot and cloudy white emulsion (positive), mandazi translucent spot (positive, cooked	Before starting, model each test for 2 minutes and say: "Goggles on, gloves on, and nobody tastes anything. Why do we use a fresh tube for each food?" Cold-call 2 groups to answer. Circulate every 5 minutes and ask: "What colour do you see? What does that tell you about this food?" Prompt: "Compare your violet with the blue control; is it really different?" Ask for water as a control (blue for biuret, no spot for the paper). WAIT TIME 10-15 seconds after asking, "What does a translucent spot mean?" before accepting answers. Ensure learners with allergies are only recording. Ask each group to put results on a class chart; cold-call 3 groups to report one result each.	Structured data collection with a class results table and controls: pooling results across groups shows patterns (beans, milk and nyama choma are positive for protein; groundnuts and mandazi are positive for lipids; ugali is negative for both) and teaches learners that tests need controls to be trusted.	Observable indicators: (1) correct safe handling (goggles, gloves, small volumes, no tasting) in at least 90% of groups; (2) results correctly recorded using the colour conventions (violet/purple = protein; translucent spot or cloudy white layer = lipid); (3) learners include a control and notice ugali is negative for protein and lipids but positive for starch; (4) the teacher checks that each group's table has all 4 foods for each test.

	in oil), milk faint translucent spot or weak emulsion, ugali negative. Learners add a 'carbohydrate' column using Lesson 3 results (ugali blue-black with iodine) to compare.			
Explain Phase  VIDEO: Introduction to amino acids <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Amino Acids (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners use their results to build a comparison table of carbohydrates, proteins and lipids: test used, positive result, building blocks (glucose units; amino acids; fatty acids and glycerol), example in Wanjiru's meal and main roles (carbohydrates: main energy source; proteins: growth and repair of body tissues, and also enzymes; lipids: energy storage and insulation). They then use the numbers (about 70 g carbohydrate in ugali plus 15 g in beans; about 20 g protein; about 10 g fat) and the graph to write an evidence-based claim: 'Carbohydrates, not proteins or lipids, are the main cause of the spike from 4.5 to 8.0 mmol/L because...' They also answer: 'Why do beans, milk and groundnuts still matter for Wanjiru even though they are not the main cause of the spike?' using growth, repair and energy storage.	Say: "Look at your tables. What evidence tells us the spike comes from ugali rather than from the oil?" WAIT TIME 10-15 seconds, then cold-call 3 learners. Draw simple structure diagrams on the board: a chain of glucose units, a chain of amino acids and a glycerol with three fatty acids, and ask: "How is a protein built differently from starch?" (cold-call 2). Ask: "Wanjiru's meal had about 70 g carbohydrate and about 10 g fat. Which would you expect to raise blood glucose more? Why?" Use the Claim-Evidence-Reasoning frame on the board. Address any misconception noted in Predict, such as 'beans cause the spike because they are sweet', by pointing to the evidence. Pair-share for 3 minutes, then have 2 pairs present claims; the class gives one question or critique per presentation. Clarify that proteins and lipids are digested too, but not mainly into glucose; the full processes come in Lessons 5 and 6, so do not teach them now.	Claim-Evidence-Reasoning with comparison: learners compare structures, tests and roles across the three classes and then justify a claim with their own data and the meal numbers, so that they link chemical differences to the phenomenon.	Observable indicators: (1) comparison table is complete and accurate (biuret/violet/amino acids; grease spot and emulsion/fatty acids and glycerol); (2) the claim cites at least 2 pieces of evidence (for example ugali negative for protein and lipid but positive for starch; 70 g carbohydrate versus 20 g protein and 10 g fat); (3) learners distinguish roles (growth and repair versus energy); (4) exit prompt: 'Name one test for protein and one for lipid and say what a positive result looks like', with at least 80% correct.
Driving Question Board (DQB) Creation  VIDEO: Peptide bond formation <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English	Learners revisit the class DQB. They place green sticky notes next to questions now answered, for example 'What chemicals are in food?' (carbohydrates, proteins, lipids, with tests), and write new questions on yellow notes, such as 'What happens to proteins in the body?', 'Are proteins digested too?', 'Do enzymes work on proteins and lipids?' and 'How	Say: "Look at the board. Which questions can we answer with evidence from today's tests?" Allow WAIT TIME of 10-15 seconds, then cold-call 3 learners to move or mark a note and justify it with evidence. Ask: "What new question has our protein and lipid evidence created?" and take at least 4 new questions. Reply to unrelated questions with: "Great	Driving Question Board as a living record: learners see their questions answered with evidence and generate new ones, which keeps the investigation learner-driven and builds ownership of the storyline.	Observable indicators: (1) at least 3 questions are moved to 'answered' with a stated piece of evidence; (2) every group adds at least 1 new, testable question; (3) at least 2 new questions refer to what happens to proteins or lipids after eating, showing connection to the phenomenon.

- US curriculum) Search ARES for similar videos EXERCISE: Amino Acid Classification and Structure Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	does glucose leave the blood?' They sort questions into: answered, to be answered in Lesson 5 (enzymes, water and minerals), and to be answered in Lesson 6 (absorption, respiration, insulin).	question; where could we find out?" Point out that enzymes are proteins, which sets up Lesson 5.		
Model Building Phase VIDEO: The Protein Folding Problem <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Protein Classification Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners revise the class flow-diagram model: Food chemicals, then digestion by enzymes, then glucose in blood, then cells (respiration) and storage, then back to normal. In groups they expand the 'Food chemicals' box to show carbohydrates (starch in ugali, about 70 g), proteins (beans, about 20 g) and lipids (cooking oil, groundnuts, mandazi, about 10 g), with the test for each and its building blocks. They draw a thick arrow from carbohydrates to 'glucose in blood' (main cause of the spike) and thin dotted arrows from proteins and lipids to 'growth, repair and energy storage'. They label question marks where they do not yet know the mechanism (enzymes, absorption, insulin). Finally they complete the summary table prompt.	Say: "Update our model with today's evidence. Where do proteins and lipids belong, and why is the arrow thinner?" Give 8 minutes in groups and use WAIT TIME 10-15 seconds before asking for explanations. Cold-call 3 groups to explain one change and the evidence for it. Ask: "What is still missing from the model that explains how the spike falls from 8.0 at 60 minutes to 4.5 mmol/L at 180 minutes?" Encourage question marks for missing mechanisms. Close with: "You didn't just learn ABOUT science—you DID science!" and ask learners to note how they feel about the process (accomplished, confident).	Cumulative model revision: adding new evidence to a shared flow diagram helps learners see how each lesson contributes to explaining the glucose graph and marks gaps for later lessons.	Observable indicators: (1) the revised model correctly shows carbohydrates as the main source of glucose and proteins and lipids as separate, thinner pathways; (2) each group cites at least 1 test result as evidence for a change; (3) groups identify at least 1 gap (for example how starch is broken down); (4) learners show confidence by explaining their model unprompted in final share.

D. TEACHER REFLECTION

Which learners confused protein and lipid test results, and did the control tubes and the comparison with carbohydrates help? Did learners use numbers (about 70 g carbohydrate, 20 g protein, 10 g fat) and the graph to argue that carbohydrates cause the spike, or did they rely on assumptions like 'beans are sweet'? Were safety routines followed, and were learners with allergies fully included? Which DQB questions about enzymes, absorption and insulin should be prioritised for Lessons 5 and 6?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that beans, milk and nyama choma turned violet/purple with the biuret test (protein), while ugali stayed pale blue. Groundnuts and mandazi left a translucent grease spot, groundnuts gave a cloudy white emulsion, and milk gave a faint result, while ugali was negative for lipids.
What did I learn?	We learned that proteins are made of amino acids and lipids are made of fatty acids and glycerol, whereas carbohydrates are made of glucose units. Proteins are used for growth and repair, lipids for energy storage, and carbohydrates are the main energy source.
How does this explain the phenomenon?	We explain that Wanjiru's blood glucose spike from 4.5 to 8.0 mmol/L at 60 minutes is mainly caused by carbohydrates (about 70 g in ugali), not by the about 20 g protein or about 10 g fat in her meal. Proteins and lipids are still used by the body, but not mainly to raise blood glucose.

LESSON 5 (80 minutes (double lesson)): Water, Mineral Salts and Enzymes in Digestion

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To investigate how saliva (salivary amylase) digests starch to reducing sugar, and how water, mineral salts and temperature affect enzyme action, so that learners can explain how the starch in Wanjiru's ugali becomes the glucose that causes her blood glucose to rise from 4.5 to 8.0 mmol/L.
Knowledge	Enzymes are biological catalysts that are proteins. Salivary amylase digests starch to maltose, which is then broken down to glucose. Digestion is the breakdown of large insoluble food molecules into small soluble ones. Water and mineral salts are inorganic compounds that support enzyme action: water is the medium in which reactions happen, and mineral salts help maintain the conditions cells need. Enzymes are denatured by boiling and inactive at 0 degrees C, and work best at about 37 degrees C in the body.
Skills	Planning and carrying out an investigation with a control (SEP 3); using iodine and Benedict's tests to collect evidence; recording and interpreting colour changes over time (SEP 4); constructing explanations from evidence (SEP 6); revising a model (SEP 2); asking questions (SEP 1).
Attitudes	Curiosity about what happens to food inside the body; respect for evidence; teamwork and responsibility when handling hot water and chemicals; appreciation that everyday food processing in our bodies is chemistry.
Key Inquiry Question	How do the chemicals of life contribute to the functioning of living organisms? (linked to: What are the chemicals of life?)
Purpose in Storyline	Lessons 3 and 4 showed that ugali is rich in starch (about 70 g carbohydrate) and that proteins differ from carbohydrates in structure and role. This lesson answers the question: how does starch become glucose that can enter the blood? Learners gather the evidence for the digestion step in the class model (food chemicals, then digestion by enzymes, then glucose in blood). Lesson 6 will use this to explain absorption, respiration, storage and insulin, and the fall back to 4.5 mmol/L.
Safety Notes	Each learner uses only their own clean, labelled cup and collects their own saliva (never share saliva; wash hands afterwards; use a clean cup per learner or have the teacher or a volunteer supply a pooled sample from consenting learners, discarded after use). Wear safety goggles or eye protection when heating Benedict's solution. Heat test tubes in a hot water bath, not in direct flame; point the tube mouth away from people; use a test tube holder. Hot water baths must be at about 37 degrees C for the incubation and boiling only under the teacher's supervision. Iodine stains skin and clothes; wipe spills immediately. Do not taste any chemicals or test materials. Dispose of mixtures as the teacher instructs and wash hands at the end.

B. LESSON OVERVIEW

Mr Otieno reminds the class of Wanjiru's blood glucose graph: 4.5 mmol/L before the meal, a peak of 8.0 mmol/L at 60 minutes after the plate of ugali, sukuma wiki and beans, then back to 4.5 mmol/L by 180 minutes. Learners already know ugali is mostly starch (Lesson 3) and that starch is too large to enter the blood directly. Learners predict what happens to starch when it meets saliva, then run the saliva and starch investigation: 2 cm³ of 1% starch solution plus 1 cm³ of saliva at about 37 degrees C, tested with iodine at 0, 5 and 10 minutes and with Benedict's at 10 minutes, alongside a control with boiled saliva or water. They explain the results using the idea of enzymes as protein catalysts, consider the roles of water and mineral salts and the effect of temperature, then add questions to the DQB and revise the class flow-diagram model to include digestion by enzymes.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Enzymes in the Digestive System (Flexbook) Source: CK-12  Search ARES for similar readings	(10 minutes) Learners look again at Wanjiru's blood glucose graph (0 min: 4.5; 15 min: 5.5; 30 min: 7.0; 60 min: 8.0 peak; 90 min: 6.5; 120 min: 5.2; 150 min: 4.7; 180 min: 4.5 mmol/L). They recall that ugali has about 70 g carbohydrate, mostly starch, and that starch gives a blue-black colour with iodine and no colour change with Benedict's before digestion. Individually they write and draw a prediction in their notebooks: 'What do you think happens to starch when it is mixed with saliva in a warm tube? What colour will the iodine test give after 0, 5 and 10 minutes? What would happen if the saliva was boiled first? Why?' They also predict whether Benedict's test will change at 10 minutes. They share in pairs and then add their ideas to a class prediction chart.	Display the graph. Say: 'Wanjiru's glucose went from 4.5 to 8.0 mmol/L in 60 minutes. But ugali is starch, not glucose. Starch is a big molecule. So what must happen to starch before it can raise glucose in the blood?' WAIT TIME 10-15 seconds in silence before taking answers. Ask: 'Have you ever noticed that if you chew ugali or bread for a long time it starts to taste slightly sweet? What might be happening in your mouth?' Cold-call 3 learners (not volunteers) to share their predictions, and ask each: 'What makes you think so?' Record without correcting. Ask: 'What do you think boiling the saliva will do?' Cold-call 2 more learners. Write all predictions on the board under headings 'Starch + saliva' and 'Starch + boiled saliva'. Say: 'Keep your prediction; today we will test it with evidence.'	Prediction with reasoning (Predict-Observe-Explain). Writing a prediction and a reason before seeing results makes learners' existing ideas about digestion visible, creates a stake in the outcome, and lets them later compare evidence with their own thinking, which deepens conceptual change. Linking prediction to the graph connects the investigation to the anchoring phenomenon.	Check notebooks as you circulate for: (1) a prediction that mentions the iodine colour changing over time; (2) a reason that links starch breakdown to sugar or glucose; (3) any mention of what saliva contains or does; (4) any prediction about the boiled saliva. Tally on a clipboard how many learners predict that the starch will disappear, that sugar will appear, and that boiled saliva will not work. Note misconceptions such as 'saliva just makes food wet' or 'the starch turns directly into blood'. Use the tally to decide how much to emphasise the control.
Observe Phase  VIDEO: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Enzymes in the Digestive System Practice Source: CK-12  Search ARES for similar readings	(30 minutes) In groups of 4, learners carry out the saliva and starch investigation. Step 1: Label tubes A (starch + saliva) and B (control: starch + boiled saliva or water). Step 2: Place 2 cm ³ of 1% starch solution in each tube. Step 3: Put a drop of iodine in each well of a white tile or spotting plate and take the 0 min sample from each tube; record the colour (both blue-black). Step 4: Add 1 cm ³ of saliva to tube A and 1 cm ³ of boiled saliva (or water) to tube B; place both in a water bath at about 37 degrees C. Step 5: At 5 min and 10 min, use a clean dropper to take a drop from each tube into	Before starting say: 'You are scientists today. Each group has a control tube. Why do you think we need tube B?' WAIT TIME 10-15 seconds, then cold-call 2 groups. Review safety: 'Own cup only, goggles on when heating, tube mouths pointing away from people.' Circulate and ask each group: 'What colour is tube A now compared with tube B? What does that tell you about the starch?' Prompt precise language: 'Say starch present or starch absent, not just a colour.' At the 10 min mark ask: 'What colour is tube A with iodine? What does that mean? Then why did Benedict's	Controlled investigation with data table (Planning and carrying out investigations; analysing data). The control tube isolates the variable (active saliva versus boiled saliva), so learners can attribute the colour changes to something in saliva rather than to warmth or time. Recording iodine at three time points shows change over time, linking evidence to the idea of progressive breakdown.	Look for: (1) learners labelling tubes correctly and using the same volumes (2 cm ³ starch, 1 cm ³ saliva) in both tubes; (2) accurate colour recording at each time point; (3) use of the control to compare; (4) correct safety behaviour (goggles, holder, tube angle); (5) learners stating that iodine changing from blue-black to yellow-brown means starch has gone and the Benedict's brick-red means reducing sugar is present. Use a checklist with a tick for each group. Ask each group to show their results table before leaving the practical.

	fresh iodine drops; record the colours (tube A: brown-black at 5 min, yellow-brown at 10 min; tube B: blue-black throughout). Step 6: At 10 min, add Benedict's solution to a sample of each tube and heat in a hot water bath for a few minutes; record the colours (tube A: brick-red; tube B: blue or no change). They complete a results table with columns for time, iodine colour (tube A and tube B), and Benedict's colour at 10 min. They also observe a teacher demonstration or discuss data on temperature: saliva and starch at 0 degrees C (in ice) stays blue-black, which shows the enzyme is inactive at 0 degrees C, and boiled saliva has no effect because the enzyme is denatured.	turn brick-red?' If groups differ from the expected colours, ask them to check timing, temperature and volume rather than telling them the answer. Ask groups to take a photo or sketch the final colours. Prompt reading of the table: 'Look at tube A from 0 to 10 min. What pattern do you see in the iodine colour?' Cold-call 3 learners to describe the pattern using evidence only. Introduce the ice-bath result or class data: 'At 0 degrees C the starch stayed blue-black. What does this tell you about the enzyme at low temperature?' Ensure all waste is collected and hands washed.		
Explain Phase  VIDEO: Enzyme cofactors and coenzymes <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(20 minutes) Learners return to their predictions and compare them with the evidence. In groups they write an explanation (claim, evidence, reasoning) for: 'How did saliva change the starch?' Claim: Saliva digested starch into reducing sugar. Evidence: iodine changed from blue-black (0 min) to brown-black (5 min) to yellow-brown (10 min), the Benedict's test at 10 min was brick-red, and the control with boiled saliva stayed blue-black. Reasoning: Saliva contains salivary amylase, an enzyme, which is a protein that acts as a biological catalyst, speeding up the breakdown of starch to maltose, which is then broken to glucose. Boiling denatured the enzyme so the starch stayed unchanged. Then learners discuss: (a) Why is the water bath at about 37 degrees C? (the temperature at which enzymes work best in the body);	Say: 'Look at your prediction. Was it supported by the evidence? Put a tick or a cross and explain.' WAIT TIME 10-15 seconds. Model the claim-evidence-reasoning frame on the board. Ask: 'What is your claim? Which evidence supports it? What science idea links the evidence to your claim?' Cold-call 4 groups to share one part each (claim, evidence, reasoning, then a link to the graph). Probe: 'Lesson 4 told us enzymes are proteins. What happened to the protein in the boiled saliva? Why did the starch stay blue-black?' WAIT TIME 10-15 seconds. Ask: 'Is the enzyme used up in the reaction? What does catalyst mean?' Clarify: a catalyst speeds up a reaction and is not used up. Ask: 'Water and mineral salts are inorganic. What role might they play in digestion?' Cold-call 2 learners, then give the key explanation. Use the term	Claim-Evidence-Reasoning (CER) with a causal explanation. CER requires learners to connect observed data (colour changes, control) to a scientific principle (enzymes as protein catalysts), which builds a mechanistic explanation. Returning to the predictions promotes reflection on changed thinking, and linking back to the graph connects the lab to the phenomenon.	Check written explanations for: (1) a claim that starch was digested to sugar; (2) at least two pieces of evidence (iodine change and Benedict's result, plus the control); (3) reasoning that names the enzyme amylase and states that it is a protein and a catalyst; (4) a correct explanation for the boiled saliva (denatured) and for 0 degrees C (inactive); (5) mention of water or mineral salts as inorganic supporting chemicals; (6) a link to the rise in Wanjiru's blood glucose. Use a 3-point rubric (claim, evidence, reasoning) and note groups needing support. Use a quick exit question: 'What does the control tell us?'

	(b) Why did the enzyme not work at 0 degrees C or after boiling? (inactive at low temperature; denatured at high temperature); (c) What roles do water and mineral salts play? (Both are inorganic, as learned in Lesson 2; water is the medium in which the starch solution and enzyme react and which carries the digested sugars, and mineral salts help keep the right conditions for enzymes and cells.) They link this to digestion: large insoluble starch becomes small soluble sugar. Finally they link to the phenomenon: starch in Wanjiru's ugali (about 70 g carbohydrate) is digested to glucose, which is then absorbed into the blood, contributing to the rise from 4.5 to 8.0 mmol/L at 60 minutes.	'digestion' (breakdown of large insoluble food molecules into small soluble ones) and avoid using 'absorption' details, which are for Lesson 6. Ask: 'Why do you think the body works best at about 37 degrees C? What would happen to the enzymes in a person with a very high fever?' Close with: 'So how does this investigation help explain the rise in Wanjiru's blood glucose?' Cold-call 3 learners, correcting misconceptions such as 'saliva turns starch into glucose instantly'.		
Driving Question Board (DQB) Creation  VIDEO: Dehydration synthesis or a condensation reaction <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Enzyme structure and catalysis Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(10 minutes) Learners revisit the class DQB with the driving question: 'What chemicals are in the food we eat, and what happens to them in our bodies?' In groups they use sticky notes to (1) mark questions that today's evidence helps to answer (for example: 'How do cells break down chemicals?' partly answered: starch is digested by enzymes to sugars; 'What chemicals are in food?' extended: water and mineral salts support digestion) and (2) write new questions that arise, such as: 'How does the digested glucose get from the gut into the blood?', 'What does the body do with glucose once it is in the blood?', 'Why does the glucose level fall back to 4.5 mmol/L?', 'Do enzymes digest proteins and fats too?', 'What happens to enzymes in a person with a high fever?' They place the notes on the DQB	Say: 'Look at the board. Which questions can we answer, even partly, with what we found today?' WAIT TIME 10-15 seconds. Invite 3 learners (cold-call) to move a note and say what evidence supports the move. Then ask: 'What new questions do you have after seeing starch turn to sugar?' Give groups 3 minutes to write at least 2 new questions each. Cold-call 3 groups to read one new question aloud and explain why it matters for Wanjiru's graph. Highlight questions that point to Lesson 6: absorption, use of glucose in cells, storage and insulin. Say: 'These are exactly the questions we will tackle next lesson.' Make sure the question 'Why does glucose not stay high forever?' is visible and starred.	Driving Question Board as a living record of learning. Sorting questions into answered, partly answered and new makes the growth in understanding visible and gives learners ownership of the inquiry, while new questions create the need to know for Lesson 6.	Observe whether learners (1) correctly identify which questions the saliva investigation addresses, (2) cite specific evidence (iodine colours, Benedict's result, control) when moving notes, (3) write new questions that are testable or point to the next missing step (absorption, respiration, insulin), and (4) link questions back to the graph. Count the number of groups that generate at least one question about what happens to glucose after digestion.

	and move answered notes to a 'Partly answered' column with a short evidence note.			
Model Building Phase  VIDEO: Competitive inhibition <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Enzymes and activation energy Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(10 minutes) Learners revise the class flow-diagram model: Food chemicals, then digestion by enzymes, then glucose in blood, then cells (respiration) and storage, then back to normal. In groups they add to the first two boxes: under 'Food chemicals' they write starch (from ugali, maize flour), protein (beans), lipids (cooking oil), water and mineral salts; under 'Digestion by enzymes' they write: amylase (a protein catalyst in saliva) digests starch to maltose, which is broken to glucose; water is the medium; works best at about 37 degrees C; denatured by boiling; inactive at 0 degrees C. They label evidence next to the arrow: 'iodine blue-black to yellow-brown; Benedict's brick-red at 10 min; boiled saliva control stays blue-black'. They add a question mark to the arrow between 'digestion' and 'glucose in blood' with the note 'absorption: Lesson 6'. Each group then completes the summary table prompt (what we observed, what we learned, how we explain it) and one group presents. The lesson ends with the celebration: learners check that they feel accomplished, value the process and are confident.	Say: 'Our model has a gap. Where in the model did today's evidence fit?' WAIT TIME 10-15 seconds. Ask groups to revise their diagram for 5 minutes. Circulate and ask: 'What evidence supports this box? What would you say to Wanjiru if she asked why her glucose rose?' Cold-call 3 groups to explain their revised model in 2 sentences using the words enzyme, catalyst, starch and glucose. Correct any use of 'simple sugar' (use 'reducing sugar' for the Benedict's test) and check that learners do not yet claim details about insulin or glycogen, which belong to Lesson 6. Update the large class model on the wall with agreed changes. Close with: 'You didn't just learn ABOUT science, you DID science!' Ask learners to tick: I feel accomplished, I value the process, I am confident. Ensure equipment is cleaned and hands washed.	Cumulative model revision (Developing and using models). Adding new evidence and mechanisms to the same flow diagram each lesson shows how scientific ideas build, and requires learners to connect parts of the system (food chemicals, enzyme digestion, glucose) to the single phenomenon. Marking gaps with question marks shows models are works in progress.	Check revised models for: (1) starch correctly placed as the food chemical digested; (2) amylase named as an enzyme, a protein and a catalyst; (3) temperature conditions (about 37 degrees C best, denatured by boiling, inactive at 0 degrees C); (4) evidence labelled on the arrow; (5) a gap identified between digestion and glucose in blood. Listen to group explanations for correct vocabulary (digestion, enzyme, catalyst, denatured, reducing sugar). Use the summary table responses as an exit check and record learners who still say that saliva is 'used up' or that starch becomes glucose without an enzyme.

D. TEACHER REFLECTION

After the lesson consider: (1) Did every group use the control tube to justify their claim about enzyme action, or did some just report colours? (2) Which learners still think that boiling or cold only 'slows' the enzyme rather than denaturing or inactivating it? (3) Was the 37 degrees C water bath stable enough for reliable results, and how could timing be improved? (4) Did learners link starch digestion back to the rise in Wanjiru's glucose from 4.5 to 8.0 mmol/L, or did they treat the investigation as separate? (5) Which new DQB questions will guide Lesson 6 (absorption, respiration, storage and insulin), and are all learners ready to explain why glucose falls back to

4.5 mmol/L? (6) Did the safety routines for saliva, hot water and iodine work well for all groups?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that with saliva at about 37 degrees C the iodine test changed from blue-black at 0 min, to brown-black at 5 min, to yellow-brown at 10 min, and Benedict's test at 10 min was brick-red, while the control with boiled saliva or water stayed blue-black.
What did I learn?	We learned that saliva contains amylase, an enzyme that is a protein and a biological catalyst, which digests starch to maltose and then glucose; water and mineral salts are inorganic chemicals that support this, and enzymes work best at about 37 degrees C, are denatured by boiling and inactive at 0 degrees C.
How does this explain the phenomenon?	We can explain that the starch in Wanjiru's ugali is digested by enzymes such as amylase into glucose, which helps explain why her blood glucose concentration rises from 4.5 to a peak of 8.0 mmol/L at 60 minutes; we still need to find out how glucose gets into the blood and why it falls back to 4.5 mmol/L by 180 minutes.

LESSON 6 (80 minutes (double lesson)): Absorption, Respiration and Regulation: Explaining the Fall Back to 4.5 mmol/L

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners complete the explanation of Wanjiru's blood glucose graph by showing what happens to glucose after digestion: it is absorbed into the blood, taken up by cells under the influence of insulin, used in respiration to release energy and stored as glycogen in the liver and muscles. Learners then finalise and present the class model that answers the driving question.
Knowledge	Learners explain that (1) digestion of starch in ugali by enzymes such as amylase produces glucose, which is absorbed into the blood, causing the rise from 4.5 to the peak of 8.0 mmol/L at 60 min; (2) the pancreas releases insulin, which causes cells, the liver and muscles to take up glucose; (3) cells use glucose in respiration (glucose + oxygen gives carbon dioxide + water + energy, ATP released); (4) the liver and muscles store excess glucose as glycogen; (5) the level falls by 3.5 mmol/L over 120 min (60 to 180 min) back to 4.5 mmol/L; (6) proteins and lipids are also digested and used for growth, repair and energy storage, but carbohydrates are the main cause of the spike.
Skills	Constructing explanations from evidence (NGSS SEP 6); developing and using models (SEP 2); analysing and interpreting data (SEP 4): calculating rates of rise and fall from the glucose table; engaging in argument from evidence (SEP 7); obtaining, evaluating and communicating information (SEP 8) through the final model presentation.
Attitudes	Learners show curiosity and persistence in solving the puzzle, respect for peers' ideas during presentations and peer critique, appreciation of how the body regulates its chemicals, and confidence in using evidence. Learners value the process of science: 'You didn't just learn ABOUT science—you DID science!'
Key Inquiry Question	1. What are the chemicals of life? 2. How do the chemicals of life contribute to the functioning of living organisms? Driving question: What chemicals are in the food we eat, and what happens to them in our bodies?
Purpose in Storyline	Lessons 1 to 5 gave the graph, the chemicals in the meal, the carbohydrate tests linking ugali starch to glucose, the roles of proteins and lipids, and the enzyme digestion of starch to sugars. Lesson 6 answers the last open question on the DQB: why the glucose does not stay high. It closes the storyline with the Final Explanation, a comparison of the Lesson 1 model with the final model, and DQB completion.
Safety Notes	This lesson uses no chemicals or heating. Learners use only the data table, role-play cards and chart paper. Keep walkways clear during the role-play. Handle food tasting only if clean, allergy checks are done and the teacher allows it (no tasting is required). Be sensitive to learners who live with diabetes or have family members who do; use the topic respectfully and do not ask learners to share health information. Ensure learners wash hands before any snack.

B. LESSON OVERVIEW

Learners return to Wanjiru's graph from Nairobi (4.5 mmol/L at 0 min, peak 8.0 at 60 min, back to 4.5 at 180 min) and finish explaining it. After predicting what happens to glucose once it is in the blood, they analyse the falling part of the graph (8.0, 6.5, 5.2, 4.7, 4.5), act out a role-play of glucose absorption, insulin release from the pancreas, uptake by cells, respiration and glycogen storage in the liver and muscles. Learners then write a Final Explanation, complete the DQB, compare the Lesson 1 model with the final flow-diagram model (Food chemicals, then digestion by enzymes, then glucose in blood, then cells (respiration) and storage, then back to normal) and present it in a gallery walk and celebration.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Cellular respiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: chemicals life absorption blood	Learners look again at Wanjiru's graph (Mr Otieno's class, Nairobi) with the table 0 min: 4.5; 15: 5.5; 30: 7.0; 60: 8.0 (peak); 90: 6.5; 120: 5.2; 150: 4.7; 180: 4.5 mmol/L. Individually (2 min) they draw on a mini whiteboard or in their notebook: 'Where does the glucose go after 60 minutes?' with a sketch of Wanjiru's body and arrows. They write one prediction: 'The glucose goes down because _____.' They also predict: 'What would the graph look like if nothing removed glucose from the blood?' and sketch a dotted line on a copy of the graph. Then they pair-share with a neighbour and note one similarity and one difference.	Display the graph and say: 'Remember Wanjiru's lunch of ugali, sukuma wiki and beans. By 60 minutes, her blood glucose is 8.0 mmol/L. Today we finish the puzzle: where does the glucose go?' Point to the falling part of the graph. Say: 'Draw your idea first, no talking.' Give 2 minutes. Then ask: 'What would happen to the graph if nothing removed glucose from the blood?' WAIT TIME 10-15 seconds before taking responses. Cold-call 4 learners to share predictions without judging, e.g. 'Akinyi, what did you draw? Why?' Record all ideas on a chart headed 'Our predictions' (e.g. 'used for energy', 'passes out in urine', 'stored', 'something controls it'). Say: 'Keep these ideas; we will test them against evidence.' Circulate and note misconceptions such as 'glucose is lost in sweat or urine'.	Predict-Observe-Explain with an initial model sketch. Making predictions visible before new evidence activates prior knowledge from Lessons 1 to 5 (the graph, glucose from starch digestion) and exposes misconceptions, so learners are motivated to look for evidence that supports or changes their ideas.	Look for: (a) at least 8 of 10 learners can sketch a falling line or arrows for glucose leaving the blood; (b) learners use ideas from earlier lessons (starch digested to glucose, enzymes); (c) identify misconceptions (glucose is just lost in urine or sweat; glucose is destroyed; glucose simply 'disappears'). Tally on a clipboard how many learners mention 'energy', 'storage' or 'a hormone'. Use this to decide how much time to spend in the Observe phase.
Observe Phase  VIDEO: Cellular respiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: chemicals life absorption blood	Part 1 (15 min): Data analysis in groups of 4. Using the glucose table, learners calculate: the rise from 4.5 to 8.0 = 3.5 mmol/L in 60 min; the fall from 8.0 to 4.5 = 3.5 mmol/L over 120 min (60 to 180 min); and the fall in each interval (60 to 90: 1.5; 90 to 120: 1.3; 120 to 150: 0.5; 150 to 180: 0.2 mmol/L). They compare: 'The rise is faster (3.5 over 60 min) than the fall (3.5 over 120 min).' They note the level is back in the normal fasting range of 4.0 to 5.5 mmol/L by 150 min (4.7). Part 2 (20 min): Role-play 'Journey of the glucose'. Groups of 8 each get role cards: glucose molecules (from ugali starch), blood (a rope loop), small intestine wall (absorption),	Say: 'Scientists look for patterns in data. Calculate the rise and the fall. Which is faster?' Move around and ask: 'How did you get 3.5 for the fall? Show me the two values.' WAIT TIME 10-15 seconds after each question. For the role-play, hand out role cards and state: 'Observers, your job is to count the glucose cards left in the blood loop after each step.' Pause the action at step 4 and ask: 'Why is the pancreas holding up insulin cards now and not at 0 minutes?' (Blood glucose is high at the peak, 8.0 mmol/L.) Cold-call 3 observers: 'How many glucose cards are left in the blood now? Which time on the graph is this?' Ask: 'Where does the energy in respiration	Data analysis plus embodied modelling (role-play). Calculating rates turns the graph into evidence of a regulating process, and acting out the invisible journey makes abstract ideas concrete (absorption, insulin, respiration, glycogen). The observers' counts link the model to the data, building the cause-and-effect reasoning needed for a scientific explanation.	Observable indicators: learners correctly calculate 3.5 mmol/L for both rise and fall and state that the fall takes twice as long as the rise (120 min versus 60 min); observers' checklists show glucose cards decreasing in step with 8.0, 6.5, 5.2, 4.7, 4.5; learners can name which role (pancreas) releases insulin and which organs store glycogen (liver and muscles). Check that learners use 'absorption' and 'respiration' correctly. Probe those who say 'insulin is food' or 'glycogen is sugar in urine'.

	pancreas (holds insulin cards), body cells (with an 'oxygen' card), liver and muscle (with 'glycogen' storage boxes). Steps: (1) amylase digestion has produced glucose (cards from Lesson 5); (2) glucose is absorbed into the blood (students step into the rope loop, blood glucose rises); (3) the pancreas releases insulin when glucose is high; (4) insulin causes cells, liver and muscles to take up glucose; (5) in cells, glucose + oxygen gives carbon dioxide + water + energy (ATP released); (6) the liver and muscles store excess glucose as glycogen; (7) blood glucose returns to 4.5 mmol/L. One observer per group records on a checklist how many glucose cards are left in the blood loop at each step and matches it to the graph (peak at 60, 6.5 at 90, 5.2 at 120, 4.7 at 150, 4.5 at 180).	come from? What gas do your cells need?' Link to breathing: 'That is why you breathe harder after running to catch a matatu.' Reinforce vocabulary: digestion (breakdown into small soluble molecules), absorption (movement into the blood). Do not re-teach enzyme action; simply recall: 'In Lesson 5, we saw amylase break starch to maltose and then glucose.' Challenge: 'Proteins and lipids from beans and cooking oil are also digested. Do they cause the spike? Use your Lesson 4 results.' (They are used for growth, repair and energy storage; carbohydrates are the main cause of the spike.)		
Explain Phase  VIDEO: ATP synthase <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: chemicals life absorption blood	Learners write the Final Explanation individually (12 min) in the Claim-Evidence-Reasoning format to answer the driving question: 'What chemicals are in the food we eat, and what happens to them in our bodies?' using Wanjiru's meal. Claim: Why does blood glucose rise and then fall? Evidence: at least four pieces of data or test results (e.g. 70 g carbohydrate in ugali, blue-black iodine for ugali; peak 8.0 mmol/L at 60 min; saliva digestion at 37 degrees C turned starch to reducing sugar, brick-red at 10 min; fall of 3.5 mmol/L over 120 min; 4.7 at 150 and 4.5 at 180). Reasoning: explain the sequence: starch is digested by enzymes such as amylase into glucose; glucose is absorbed into the blood so concentration rises to 8.0; the	Say: 'You are now the scientist. Explain the whole story from the plate of ugali to the graph returning to normal.' Show the CER frame on the board with sentence starters: 'The data show that...', 'This is because...', 'Insulin causes...'. WAIT TIME 10-15 seconds after posing: 'Which piece of evidence best shows that the glucose came from the starch in ugali?' Cold-call 3 learners for evidence and 2 for reasoning. Circulate with a checklist: Does the explanation include (1) digestion by enzymes, (2) absorption, (3) insulin from the pancreas, (4) respiration with the equation, (5) glycogen in the liver and muscles, (6) numbers from the table? Prompt gaps: 'You wrote that glucose disappears. Where does it go? What chemical equation	Claim-Evidence-Reasoning (CER) with peer critique. Requiring a claim linked to specific data and a chain of reasoning shifts learners from recalling facts to constructing a causal explanation, integrating evidence from all five lessons. Peer review makes reasoning visible and improves it.	Use a 4-level rubric on the written explanation: Level 4 includes all six components with correct data (3.5 mmol/L rise in 60 min, 3.5 mmol/L fall over 120 min, 8.0 peak, 4.5 at 180 min); Level 3 lacks one component; Level 2 describes the graph only; Level 1 lists facts without linking. Look for correct use of 'blood glucose concentration', 'absorption', 'respiration', 'insulin' and 'glycogen'. Target: at least 70% of learners reach Level 3 or 4. Note learners who mix up digestion and absorption or say insulin is a food, and reteach in the next session.

	pancreas releases insulin; cells, liver and muscles take up glucose; cells use it in respiration to release energy (glucose + oxygen gives carbon dioxide + water + energy, ATP released); liver and muscles store the excess as glycogen; the blood level falls back to 4.5 mmol/L by 180 min. They also state the roles of proteins (beans, growth and repair) and lipids (cooking oil, energy storage) and why carbohydrates are the main cause of the spike. Then, in groups of 4, they peer-review using a checklist and revise one sentence.	shows what cells do with it?' Ask a challenge: 'If the pancreas released no insulin, how would the graph after 60 minutes look different?' (Level would stay high; do not extend to diseases beyond what is needed, just say regulation would fail.) After peer review, ask 3 groups to read one reasoning sentence aloud and invite the class to give a thumbs up or ask for more evidence. Conclude: 'Notice that carbohydrates are the main cause of the spike, but proteins and lipids from beans, groundnuts and mandazi are also digested and used for growth, repair and energy storage.'		
Driving Question Board (DQB) Creation  VIDEO: Cellular respiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: chemicals life absorption blood	Learners revisit the class Driving Question Board started in Lesson 1, headed 'What chemicals are in the food we eat, and what happens to them in our bodies?' In groups, they take the sticky notes of questions and sort them into 'Answered', 'Partly answered' and 'New questions'. For each answered question, they attach an evidence note naming the lesson where it was answered (e.g. 'Why does glucose rise?' answered by Lessons 3 and 5; 'What chemicals are in food?' answered by Lessons 2, 3 and 4; 'Why doesn't glucose stay high forever?' answered today). Each learner writes one new question on a different-coloured sticky note (e.g. 'How do the kidneys or hormones other than insulin affect glucose?' or 'How does exercise change the graph?' or 'How would the graph differ after eating mandazi?') and places it in the 'New questions' section.	Say: 'Let us go back to our first questions from Lesson 1. Which can we now answer with evidence?' Read 3 original questions aloud and ask: 'Who can answer this using data or a test we did?' WAIT TIME 10-15 seconds. Cold-call 5 learners to move a question to 'Answered' and justify it with a lesson and evidence. Ask: 'Which questions are only partly answered? What else would we need to find out?' Highlight that real scientists end with new questions. Say: 'Write one question that Mr Otieno could not answer from our data.' Have learners place their notes and select 2 for a 'future investigation' list (e.g. effect of exercise on the graph; comparing a meal of mandazi with one of ugali). Link back to the KICD key inquiry questions: 'What are the chemicals of life? How do the chemicals of life contribute to the functioning of living organisms?' and ask learners to point to DQB notes that answer each.	Question sorting on the Driving Question Board. Revisiting the learners' own questions shows the growth of knowledge across the unit, gives ownership of the learning and models how science generates new questions from answered ones.	Observable indicators: learners correctly match at least 80% of questions to the lesson and evidence that answered them; they can explain why a question is only 'partly answered'; new questions are testable and related to the chemicals of food or glucose regulation (not random). Count how many learners can quote a specific data value (e.g. 70 g carbohydrate, 8.0 mmol/L, 37 degrees C) when moving a note.

Model Building Phase  VIDEO: Cellular respiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: chemicals life absorption blood	Part 1 (10 min): Learners retrieve their Lesson 1 model (their first sketch) and the class model. In groups of 4, they build the final flow-diagram model on chart paper: Food chemicals (carbohydrates, proteins, lipids, vitamins, water, mineral salts in ugali, sukuma wiki and beans), then digestion by enzymes (starch to maltose to glucose by amylase), then glucose in blood (absorption; peak 8.0 mmol/L at 60 min), then cells (respiration: glucose + oxygen gives carbon dioxide + water + energy, ATP released) and storage (liver and muscles store glycogen; insulin from the pancreas causes uptake), then back to normal (4.5 mmol/L by 180 min). They label the graph points (8.0, 6.5, 5.2, 4.7, 4.5) on a small graph beside the model and add the roles of proteins (growth and repair) and lipids (energy storage). Part 2 (15 min): Gallery walk and presentation. Two groups present for 2 min each; others visit posters and leave a 'star' (strength) and a 'step' (suggestion) sticky note. Part 3 (5 min): Model comparison. Learners place their Lesson 1 model beside the final model and complete 'At first we thought... Now we know... because of evidence from...' Closing celebration: learners tick 'I feel accomplished', 'I value the process' and 'I am confident' and the teacher says, 'You didn't just learn ABOUT science—you DID science!'	Say: 'Your model must answer the driving question and explain every part of Wanjiru's graph. Where does the rise come from? Where does the fall come from?' Provide chart paper, markers and a checklist: includes digestion, absorption, insulin, respiration with equation, glycogen, data labels, arrows showing the return to normal. WAIT TIME 10-15 seconds after asking: 'What is missing from your model that would explain the fall after 60 minutes?' Circulate and ask each group: 'Show me where insulin acts. What happens to the glucose after it enters the cell?' During the gallery walk, cold-call 4 learners to give one 'star' and one 'step' to a group. Ask the class: 'How is this model different from the one we drew in Lesson 1?' Cold-call 3 learners to name one idea they added (e.g. enzymes, insulin, glycogen, respiration). Ask presenters: 'What evidence in your model came from our own tests?' Connect to the careers of nutritionists and doctors in Nairobi and Mombasa hospitals, and to farmers like Mama Achieng near Kisumu whose maize and beans provide these chemicals. Close with the celebration: 'Look at how far your thinking has come, from a puzzling graph to a full explanation. You didn't just learn ABOUT science—you DID science!' Invite 3 learners to say one thing they are proud of.	Cumulative model revision and comparison (first model versus final model), with gallery walk and peer feedback. Comparing the first and final models makes learning visible and builds the ability to see a system as a whole: food chemicals, digestion, absorption, regulation and use of energy. Peer feedback develops scientific communication and argument from evidence.	Use a model checklist: (1) all five stages in the correct order (food chemicals, digestion by enzymes, glucose in blood, cells and storage, back to normal); (2) insulin from the pancreas shown as lowering blood glucose; (3) respiration equation correct; (4) glycogen stored in the liver and muscles; (5) data labels match the table (peak 8.0 at 60, 6.5 at 90, 5.2 at 120, 4.7 at 150, 4.5 at 180); (6) proteins and lipids roles stated. Success: at least 80% of groups meet 5 of 6 criteria. Listen to presentations for correct use of key terms. Learners' 'At first we thought... Now we know...' statements should name at least two new ideas supported by evidence.
--	---	---	--	--

D. TEACHER REFLECTION

After the lesson, reflect: (1) How many learners could explain the fall in glucose using insulin, respiration and glycogen, and which parts of the CER explanations were

weakest? (2) Did the role-play help learners connect the model to the graph, or did some treat it as a game without the data link? (3) Which misconceptions remained (for example, glucose lost in urine or sweat, insulin as food, confusing digestion with absorption)? (4) Did every learner contribute to the DQB and the model presentation, including quieter learners? (5) Which learners could compare their Lesson 1 model with the final model with specific evidence? (6) What would I change, such as time for the gallery walk or support for learners who struggle with the respiration equation? Record two follow-up actions, such as a short revision quiz and an extension on how exercise changes the glucose graph.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that after Wanjiru's meal of ugali, sukuma wiki and beans, blood glucose concentration rose from 4.5 to a peak of 8.0 mmol/L at 60 min (a rise of 3.5 mmol/L) and then fell by 3.5 mmol/L over the next 120 min, to 6.5 at 90, 5.2 at 120, 4.7 at 150 and 4.5 mmol/L at 180 min. The rise is faster than the fall. In our role-play and data analysis we saw glucose enter the blood and then leave it.
What did I learn?	We learned that starch in ugali is digested by enzymes such as amylase into glucose, which is absorbed into the blood. The pancreas releases insulin, which causes cells, the liver and muscles to take up glucose. Cells use glucose in respiration (glucose + oxygen gives carbon dioxide + water + energy, ATP released) and the liver and muscles store the excess as glycogen. Proteins and lipids are also digested and used for growth, repair and energy storage.
How does this explain the phenomenon?	Blood glucose rises after the meal because starch and other carbohydrates in the food are digested into glucose and absorbed into the blood. It falls back to 4.5 mmol/L by 180 min because insulin from the pancreas causes glucose to be taken up by cells, used in respiration to release energy, and stored as glycogen in the liver and muscles. Carbohydrates are the main cause of the spike, while proteins and lipids from beans, milk, groundnuts and mandazi are used for growth, repair and energy storage. This answers our driving question about the chemicals in the food we eat and what happens to them in our bodies.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.